Chapter 1

Standard Model of Cosmology

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Introduction

Cosmology is the study of the Universe, aiming to understand its origin and evolution. Within the broader landscape of science and physics, it occupies a unique position: it is one of the few fields where phenomena cannot be reproduced to directly test theoretical predictions. We have access to a single realisation of the Universe, cannot replicate its underlying mechanisms, and can only observe a limited portion of its information.

This thesis aims to contribute to our understanding of the earliest moments of the Universe. This first chapter provides a structured overview of the standard cosmological model, known as ΛCDM, guiding the reader from its fundamental principles to its current limitations. The chapter is organized into five sections. The first introduces the key components of the model, including its foundations, assumptions, and limitations, while the second presents a qualitative overview of the thermal history of the Universe. The third focuses on the Cosmic Microwave Background (CMB), one of the most powerful probes in observational cosmology, and in particular of the early Universe. The fourth section presents the main sources of contamination affecting CMB observations. Finally, the fifth section reviews past CMB experiments and their major results, before discussing upcoming missions.

1.1 The Standard Model of Cosmology

Modern cosmology is built upon a set of simple but powerful principles that allow us to describe the large-scale behaviour of the Universe. Observations reveal that the Universe is expanding and appears remarkably homogeneous and isotropic on large scales. These properties, alongside with General Relativity provide the foundation for the standard cosmological model, known as ΛCDM.

In this section, we introduce the key elements of this framework. We begin with the observational evidence for the expansion of the Universe, then present the assumptions underlying its large-scale description. We subsequently describe the main components of the Universe, before discussing the limitations of this model and the role of inflation as a possible resolution.

1.1.1 Basics of General Relativity

General Relativity (GR), developed by Einstein in 1915 [1], extends the principles of Special Relativity to non-inertial frames and gravitation. In Special Relativity, the laws of physics are invariant in all inertial reference frames, and the speed of light in vacuum, noted c, is constant for all observers. This implies a fundamental causal structure of spacetime, in which no information or physical object can propagate faster than light:

$$v < c.$$

This principle ensures causality, meaning that cause always precedes effect in every physical process.

In Special Relativity, spacetime is described by the Minkowski metric:

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \tag{1.1}$$

which defines the invariant spacetime interval:

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu.$$

This metric describes a flat spacetime in the absence of gravitation.

General Relativity generalizes this framework by introducing a curved spacetime described by a metric tensor $g_{\mu\nu}$:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

In this theory, gravity is no longer interpreted as a force but as a manifestation of spacetime curvature produced by matter and energy. The dynamics of spacetime are governed by the Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

where $G_{\mu\nu}$ describes the geometry of spacetime, $T_{\mu\nu}$ the matter and energy content, G the gravitational constant, and Λ the cosmological constant.

Particles and light propagate along trajectories determined by the geometry of spacetime. In particular, light follows null geodesics satisfying:

$$ds^2 = 0,$$

which preserves the universal causal structure inherited from Special Relativity.

1.1.2 Expanding Universe

A major milestone in modern cosmology was the discovery that the Universe is expanding, independently established by Lemaître in 1927 [2] and Hubble in 1929 [3], based on galaxies observations from Vesto Slipher between 1912 and 1917 [4, 5]. Observations showed that distant galaxies apparently recede from each other, with a velocity proportional to their distance. The proportional factor is called *Hubble constant*, and can be seen in Figure 1.1. This relation, now known as Hubble-Lemaître law, provided the first evidence that the Universe is not static but in global expansion.

This expansion is described by the *scale factor* $a(t)$, which quantifies how distances between comoving points evolve with time. By convention, the scale factor is normalised to $a(t_0) = 1$ today. As the Universe expands, the scale factor increases, implying that physical distances between distant, gravitationally unbound objects grow proportionally with time The expansion rate is quantified by the Hubble parameter:

$$H(t) = \frac{\dot{a}(t)}{a(t)},$$

with *Hubble constant* being $H(t_0) \equiv H_0$.

An observable consequence of this expansion is the *cosmological redshift* z. As light propagates through an expanding Universe, its wavelength is stretched along with space. This effect is not a simple Doppler redshift [6], but rather a consequence of the expansion of spacetime, hence the name of cosmological redshift, illustrated in

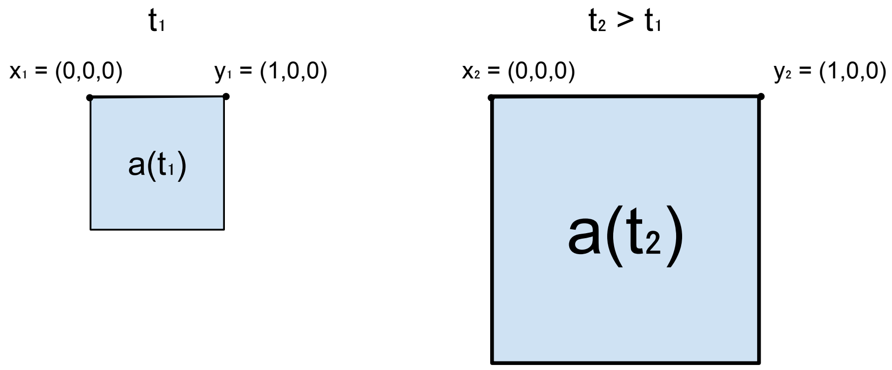

Figure 1.1: Illustration showing the difference between comoving and physical distances. At a time t_1, two points x_1 and y_1 are separated by a distance 1. After some time, at t_2, the Universe has expanded: the scale factor increased and the physical distance between x and y is larger proportionally to the scale factor. In comoving coordinates, however, the distance between x_2 and y_2 remains unchanged.

Figure 1.2. Nevertheless, the interpretation of cosmological redshift is misleading, as it has nothing to do with any recession velocity. It is often interpreted as the stretching of space due to expansion, which would have the effect of increasing the wavelength along photon's trajectory. However, general relativity asserts that local reference frame can't be distinguished from Minkowski space, and by extension, that the photon will never know about spacetime stretching. The observed shift in photon's frequency is rather due to integration of infinitesimal Doppler shifts along the travel path, caused by Lorentz' boost to go from one Minkowski frame to another. (see [7] for more accurate description). Yet, the redshift can still be simply defined as:

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} \equiv \frac{a_{\text{obs}}}{a_{\text{emit}}} = \frac{1}{a_{\text{emit}}} \tag{1.2}$$

This relation allows us to infer distances and the expansion history of the Universe from observations of spectral lines. In his original work, Hubble measured the redshift of galaxies and compared it to their distances, establishing the linear relation between velocity and distance shown in Figure 1.3. This observation demonstrates that the expansion of the Universe is uniform on large scales: every observer sees distant galaxies receding, without implying a special position in space.

1.1.3 Cosmological principle

The fundamental concept of the cosmological principle states that the Universe is homogeneous and isotropic at sufficiently large scales. This assumption considerably simplifies the Einstein field equations [1] and forms the basis of modern cosmology. Einstein first applied this principle in 1917 to construct a static cosmological model [9]. However, as discussed in Section 1.1.2, the hypothesis of a static Universe became incompatible with later observational evidence of cosmic expansion obtained by Lemaître and Hubble.

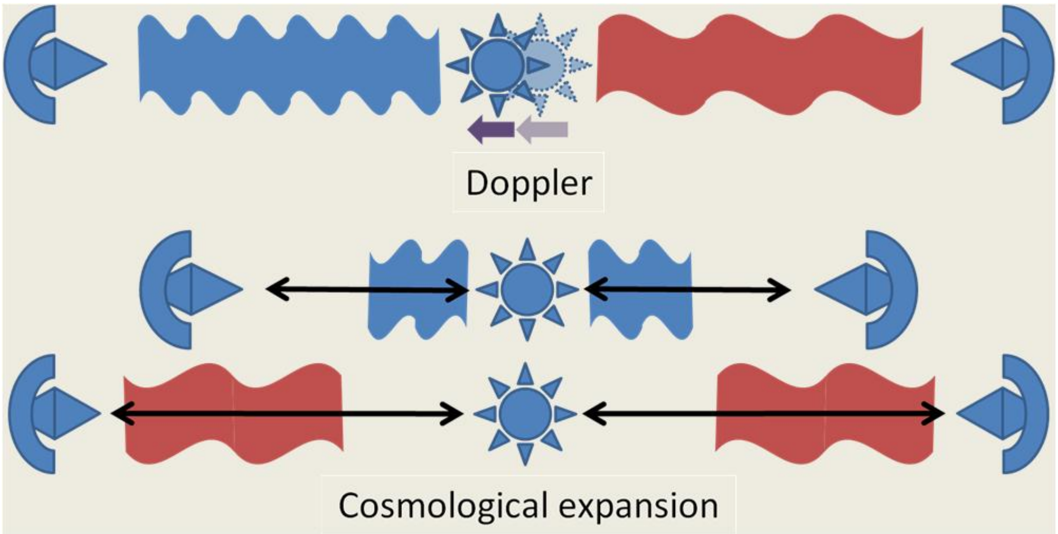

Figure 1.2: Illustration, from [8], showing the difference between Doppler redshift and cosmological redshift. Doppler redshift [6], on top, is the decrease in wavelength of an object approaching us, and the broadening of an object moving away from us. Cosmological redshift [3] is the broadening of the wavelength in all directions due to the expansion of the Universe.

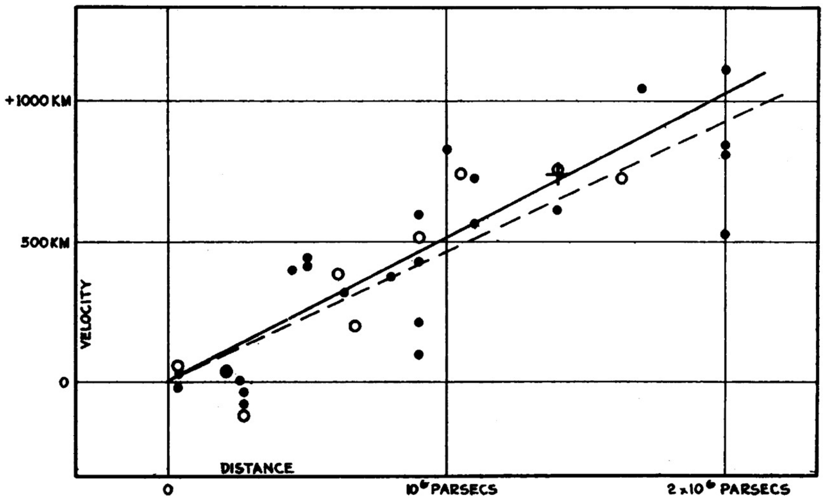

Figure 1.3: Historical velocity-distance diagram from Edwin Hubble [3]. It shows the recession velocity of galaxies as a function of their distance, illustrating the linear relation now known as Hubble's law.

In 1922, Friedmann demonstrated that Einstein's equations naturally admit homogeneous and isotropic non-static solutions [10], quickly followed by Lemaître's interpretation of the observed galactic redshifts as evidence for cosmic expansion in 1927 [2]. In parallel to these developments, alternative solutions of Einstein's equations also highlighted the role of global structure in determining local physics. In particular, the de Sitter Universe [11], obtained in the absence of matter, describes an expanding spacetime driven solely by the cosmological constant, showing that cosmic expansion can arise even without a material source.

These ideas are closely connected to Mach's principle[1], according to which local inertial properties should be determined by the global distribution of matter in the Universe. In this view, rotational motion is not defined with respect to an absolute space, as in Newtonian mechanics, but relative to the large-scale matter distribution of the Universe. Einstein was strongly influenced by this principle during the development of General Relativity, where no preferred inertial frame exists a priori. Nevertheless, General Relativity does not fully implement Mach's principle, since vacuum solutions such as de Sitter spacetime remain valid even in the absence of matter.

Building on these results, Robertson [13] and Walker [14] formulated the general relativistic metric describing a homogeneous and isotropic expanding Universe. This metric, now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, forms the cornerstone of the ΛCDM cosmological model. It is written as:

$$ds^2 = c^2 dt^2 - a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \tag{1.3}$$

where $a(t)$ is the *scale factor*, introduced in Section 1.1.2, and k is the spatial curvature parameter. The latter determines the geometry of spatial hypersurfaces:

$$k = \begin{cases} +1 & \text{closed Universe (positive curvature),} \\ 0 & \text{flat Universe (Euclidean space),} \\ -1 & \text{open Universe (negative curvature).} \end{cases}$$

By inserting the FLRW metric into Einstein's field equations, one obtains the Friedmann equations, which govern the evolution of the scale factor depending on the content of the Universe (see 1.1.4):

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{1.4}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3P}{c^2} \right) + \frac{\Lambda c^2}{3}, \tag{1.5}$$

where ρ and P are respectively the total energy density and pressure of the cosmological fluid.

The first Friedmann equation relates the expansion rate of the Universe to its matter-energy content, spatial curvature, and cosmological constant. The second equation describes the acceleration of the expansion. In particular, ordinary matter and

[1] The term "Mach's principle" was introduced by Einstein based on the ideas developed by Mach in [12].

radiation contribute to slowing down the expansion through gravitational attraction, while a positive cosmological constant leads to accelerated expansion. These equations provide the theoretical framework describing the successive eras of cosmological evolution, including radiation domination, matter domination, and the current epoch of dark-energy domination. Their solutions determine the thermal and dynamical history of the Universe within the standard cosmological model.

Although the cosmological principle is fundamentally an assumption, modern observations strongly support its validity on large scales. In particular, measurements of the Cosmic Microwave Background (CMB) by the Planck satellite show that the temperature anisotropies are of order 10^{-5} above angular scales of approximately 6° [15, 16], indicating that the Universe is remarkably homogeneous and isotropic at large scales.

1.1.4 Content of the Universe

In the two previous subsections 1.1.2 and 1.1.3, we introduced the standard description of the Universe: it is homogeneous, isotropic and expanding. We now provide a brief overview of its contents. These contents can be classified into three categories: radiation, matter, and the hypothetical cosmological constant. Each of these components evolves differently with the scale factor that describes the expansion of the Universe. Moreover, each component dominated the Universe at different epochs in its history. The evolution of each density is shown in Figure 1.4.

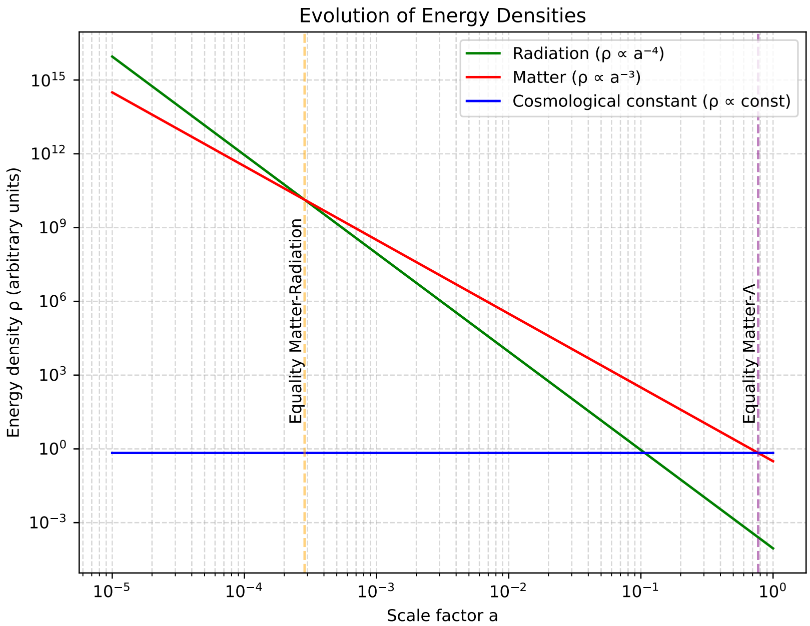

Figure 1.4: Energy density of the Universe's components as a function of the scale factor, highlighting the epochs when matter equals radiation and when matter equals the cosmological constant.

At early times of the Universe, the radiation was dominating. This radiation consisted of the relativistic particles: photons and neutrinos. At this time, the Universe was a very dense plasma of protons and electrons at thermal equilibrium, where photons' free path was very small, forming a nearly perfect blackbody. The density of radiation and matter was equal at $z \sim 3400$ [17, 18], which corresponds to approximately 80000 years after first times of the Universe. Today, radiation contributes about 0.008% of the total energy density, as inferred from Planck 2018 data [18].

As the Universe expanded, non-relativistic matter came to dominate the energy density. This matter is composed of baryonic matter and dark matter. The term "baryonic matter" is misleading, as it designates all non-relativistic particles of the Standard Model; however, its energy density is overwhelmingly dominated by protons and neutrons, hence the terminology. On the other hand, dark matter is a hypothetical form of matter, which only (or mostly) interacts through gravitational force, and thus, which does not emit light. It was first suggested by comparing "seen mass" computed from galaxy brightness and "unseen mass" computed from galaxy velocities in the Coma Galaxies Cluster by Zwicky in 1933 [19]. The mass computed from the velocity of galaxies was much higher than the luminous mass, suggesting the presence of massive matter that does not emit light.

Since then, multiple independent observations have provided strong evidence for dark matter: galaxy rotation curves [20], the Bullet Cluster through gravitational lensing [21], Baryon Acoustic Oscillations peaks observed in CMB [18] and Large Scale Structure (LSS) [22, 23], formation rate of LSS [24], and many others. While we still don't know its nature from the particle point of view, its cosmological properties are well constrained. It is commonly divided into three categories: cold, warm and hot dark matter, which refer to velocity and mass (or equivalently, whether it was relativistic in the early Universe). This classification determines how far dark matter can propagate (free-stream) and thus influences the formation of structures at different scales. Observations strongly favour the Cold Dark Matter (CDM) scenario [25], in which dark matter is non-relativistic. Warmer dark matter would have required larger primordial perturbations than measured to form observed LSS.

Today, Cold Dark Matter (CDM) represents about 26.5% [18] of total energy density, and plays a central role in Standard Model of Cosmology, ΛCDM. The ordinary matter is responsible for 4.5% [18] of the energy density budget, meaning that the energy density of matter occupies 31% of the energy density of the Universe, and that 85% of matter in the Universe is composed of dark matter.

At $z \sim 0.3$, corresponding to an age of about 10 Gyr, the Universe transitions from matter domination to dark-energy domination, as the dark-energy component becomes the dominant contribution to the total energy density. The concept of dark energy was introduced to explain the observed acceleration of the expansion of the late Universe from distant Type Ia supernovae in 1998 [26, 27]. In the standard cosmological model, this accelerated expansion can be described by introducing a cosmological constant Λ [28, 29], which corresponds to a vacuum energy density that is constant in time and space. However, dark energy is defined more generally as any component with negative pressure responsible for the acceleration, and the cosmological constant is only its simplest possible realisation. Today, dark energy remains one of the greatest unknowns in cosmology and fundamental physics. It accounts for approximately 69% of the total energy density of the Universe and is the dominant component of its current energy budget [30].

1.1.5 Problems of the Standard Model

From the previous subsection, the Standard Model of cosmology describes the Universe as homogeneous, isotropic, expanding, and composed of radiation, matter and dark energy. It is enough to describe the evolution of the Universe, and match remarkably well with observations. However, fundamental questions remain unanswered, and in particular three issues cannot be explained within this framework: the horizon problem, the flatness problem, and the monopole problem.

Horizon problem: Why is the CMB temperature so uniform ? As discussed previously, at the time of recombination the Universe was extremely homogeneous, with density fluctuations of order 10^{-5}, and consequently the cosmic microwave background (CMB) exhibits an almost perfectly uniform temperature. This uniformity is naturally explained if the early Universe was in thermal equilibrium, consistent with the fact that the CMB spectrum is an almost perfect blackbody [31]. However, within the standard cosmological model, regions of the CMB that are widely separated on the sky were never in causal contact: light signals would not have had enough time to travel between them since the beginning of the expansion. These regions therefore lie outside each other's particle horizon and cannot have exchanged information or thermalised. This raises a fundamental question: why do causally disconnected regions have the same temperature?

Flatness problem: Why is the Universe flat ? In general relativity [1], the geometry of spacetime is determined by its energy content. In a homogeneous and isotropic Universe, this relation is encoded in the Friedmann equations [10], which relate the expansion rate to the total energy density and spatial curvature. From these equations, one defines the critical density, ρ_c, which corresponds to a spatially flat Universe. Depending on the ratio $\Omega = \rho/\rho_c$, the Universe can be:

$$\begin{aligned}
\Omega = 1 &\quad : \text{flat (Euclidean geometry)}, \\
\Omega > 1 &\quad : \text{closed (positive curvature)}, \\
\Omega < 1 &\quad : \text{open (negative curvature)}.
\end{aligned}$$

Within the standard Friedmann evolution, flatness is not a stable state: any small deviation from $\Omega = 1$ grows with time. However, current measurements, notably from the Planck mission [18], indicate that Ω differs from unity by less than about 0.1%, implying that the Universe is extremely close to spatial flatness. Consequently, evolving backward in time implies an extreme fine-tuning of the initial condition. For example, a deviation of order $|\Omega_0 - 1| \sim 10^{-3}$ today corresponds to an initial deviation of order 10^{-60} near the Planck epoch. This extraordinary sensitivity suggests an apparent fine-tuning problem: why was the early Universe so precisely close to flatness?

Monopole problem: Why are magnetic monopoles not observed ? Grand unified theories (GUTs), which attempt to unify the electromagnetic, weak, and strong interactions at high energy scales, generally predict the existence of topological defects formed during phase transitions in the early Universe. Among these relics are magnetic monopoles, hypothetical particles carrying an isolated magnetic charge, whose existence was first proposed by Dirac in 1931 [32]. In standard cosmology, such monopoles should be efficiently produced during the spontaneous symmetry breaking associated with the GUT phase transition. Causally disconnected regions of space choose different vacuum states, leading to the formation of stable topological defects. As a consequence, one expects a large relic abundance of monopoles. However, monopoles are not observed in the present Universe, despite extensive experimental searches. This discrepancy constitutes the monopole problem: standard cosmology predicts a relic density of monopoles that is many orders of magnitude larger than observational bounds.

An answer to the three problems was proposed by Alexei Starobinsky in 1979 [33] and Alan Guth in 1981 [34], through the idea of an exponential expansion phase at the very beginning of the Universe. The model was then developed by Andrei Linde in 1983 [35], and has since given rise to a wide variety of models. We will explain in the following subsection the idea behind this theory and how it can solve the issues discussed previously.

1.1.6 Inflation theory

The theory of Inflation postulates that in the very early Universe, all regions were causally connected. These regions were then disconnected due to an exponentially expanded phase. This theory was developed to solve the various problems presented previously 1.1.5.

Horizon Problem. Indeed, if all regions were connected in the very early Universe, they could have thermalised, explaining why the CMB temperature is uniform at a very large scale.

Flatness Problem. A nearly flat Universe is an unstable solution, meaning that it is very unlikely to happen. But, Inflation suggests that they do not care about the curvature before it occurs, as an exponential expansion will make all local curvatures negligible. We can understand this argument with the analogy of a sphere: if we zoom enough on a sphere, its surface will appear flat. This mechanism explains why the Universe is flat today, independently of its initial conditions.

Monopole Problem. The GUT theories predict the formation of monopoles, but the exponential expansion dilutes their density to the point where they become unobservable today.

An additional issue arises within the equations of the standard cosmological model. As for any dynamical system, the evolution of the Universe is fully determined only once a set of initial conditions is specified. This naturally raises the question of how such initial conditions can be justified or constrained observationally. In the context of the ΛCDM model, the observed properties of the Universe (large-scale homogeneity, isotropy, and near spatial flatness) must be imposed as initial conditions. From a theoretical perspective, these conditions may appear finely tuned, since they require the early Universe to occupy a very specific region of phase space. The inflationary paradigm provides a dynamical mechanism to address this issue. A period of accelerated, approximately exponential expansion drives the Universe toward homogeneity, isotropy, and flatness, effectively suppressing sensitivity to the initial state. In this sense, inflation acts as an attractor mechanism in the space of initial conditions. Moreover, inflation naturally generates primordial density perturbations. Quantum fluctuations of the inflaton field are stretched to cosmological scales by the rapid expansion, where they become classical perturbations. These perturbations seed the anisotropies observed in the Cosmic Microwave Background (CMB) and later grow via gravitational instability to form the large-scale structure of the Universe observed today [36].

The most widely studied inflationary scenario involves a scalar field that slowly evolves toward its true ground state [37] [38]. In this framework, the field's potential energy remains nearly constant and therefore rapidly comes to dominate over both its kinetic energy and the energy density of other components. Inflation comes to an end when the field reaches the minimum of its potential, after which it undergoes oscillations and decays into lighter particles, eventually producing those of the Standard Model of Particles. A wide variety of potential shapes have been proposed in the literature (see [39] for an overview of the different models), and many of these can be adjusted to fit current observations. This highlights the importance of observational constraints in discriminating between competing inflationary models.

1.2 Thermal history of the Universe

In this section, we aim to produce a quick overview of the history of the Universe, discussing the key milestones of its evolution and their implication.

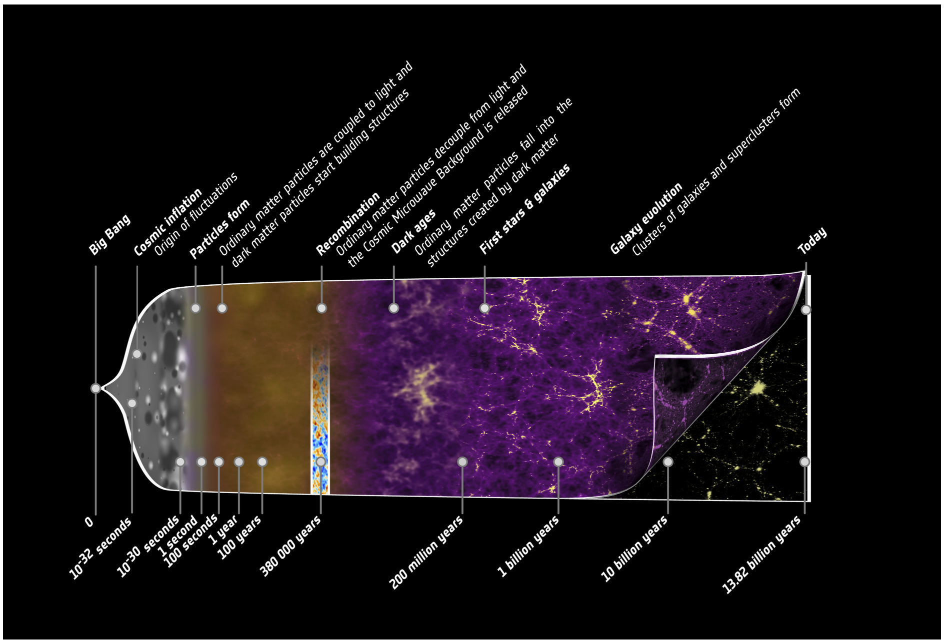

Figure 1.5: Illustration of the history of the Universe from the early beginning to today, highlighting the key events discuss in this section. Credit: ESA – C. Carreau.

1.2.1 Inflation and First particles

As discussed in Section 1.1.6, the standard cosmological model predicts an early phase of accelerated, quasi-exponential expansion known as inflation, occurring at energies of order 10^{16} GeV (corresponding to $t \sim 10^{-36} - 10^{-32}$ s). This paradigm addresses several major issues in cosmology and provides a mechanism for setting the initial conditions of the observable Universe. At the end of inflation, the scalar field driving this expansion reaches the minimum of its potential and decays into relativistic particles as the Universe cools. This process, known as reheating, occurs at characteristic energy scales of order $10^9 - 10^{15}$ GeV and marks the onset of a radiation-dominated hot plasma. Subsequently, at temperatures ranging from $T \sim 10^{12}$ GeV down to the electroweak scale $T \sim 100$ GeV, matter and antimatter are produced and remain in thermal equilibrium. During this epoch, baryogenesis occurs at an unknown high-energy scale, hypothetically generating a baryon asymmetry of order 10^{-10}, which prevents complete matter-antimatter annihilation. As the Universe expands and cools further, at the electroweak phase transition ($T \sim 100$ GeV), the electroweak symmetry is broken, separating the electromagnetic and weak interactions. At lower energies ($T \sim 1$ GeV), the QCD confinement transition occurs, leading to the formation of hadrons, primarily protons and neutrons, as quarks become confined into colour-neutral states. At temperatures around $T \sim 1$ MeV ($t \sim 1$ s), neutrinos decouple from the primordial plasma, forming a relic background known as the cosmic neutrino background. Shortly afterward, electron-positron pairs annihilate efficiently when the temperature drops below the electron mass scale ($T \lesssim 0.5$ MeV), transferring their entropy to photons. This sequence of symmetry-breaking transitions are well described by the Standard Model of Particles Physics [40, 41].

The resulting Universe is dominated by radiation (photons and neutrinos) with a small residual baryonic matter component, which later seeds the formation of large-scale structure [42].

1.2.2 Big Bang Nucleosynthesis

At early times, the temperature of the Universe was too high for nuclei to form, as any bound state was immediately broken by high-energy photons. As the Universe expanded and cooled, this process became inefficient, allowing nuclear reactions to proceed.

Weak interactions froze out at a temperature of $T \sim 0.8$ MeV ($t \sim 1$ s), fixing the neutron-to-proton ratio at $n/p \sim 1/6$, which subsequently decreased due to free neutron decay of $\sim 880s$. Big Bang Nucleosynthesis (BBN), first introduced in [43], effectively began when the temperature dropped below $T \sim 0.1$ MeV ($t \sim 200$ s), enabling the formation of deuterium through

$$p + n \to {}^2\mathrm{H} + \gamma \,. \tag{1.6}$$

This marked the end of the so-called *deuterium bottleneck*. Rapid nuclear reactions then led to the synthesis of light elements, primarily ^{3}He, ^{4}He, and trace amounts of ^{7}Li. Due to its large binding energy, ^{4}He is the most stable and abundant product after hydrogen, with a final mass fraction of about 25%, while the remaining baryonic matter consists mostly of hydrogen.

The predicted primordial abundances are in good agreement with observations, making BBN one of the key pillars of the standard cosmological model [44]. The Figure 1.6 shows the evolution of abundances of the lighter elements with time and energy.

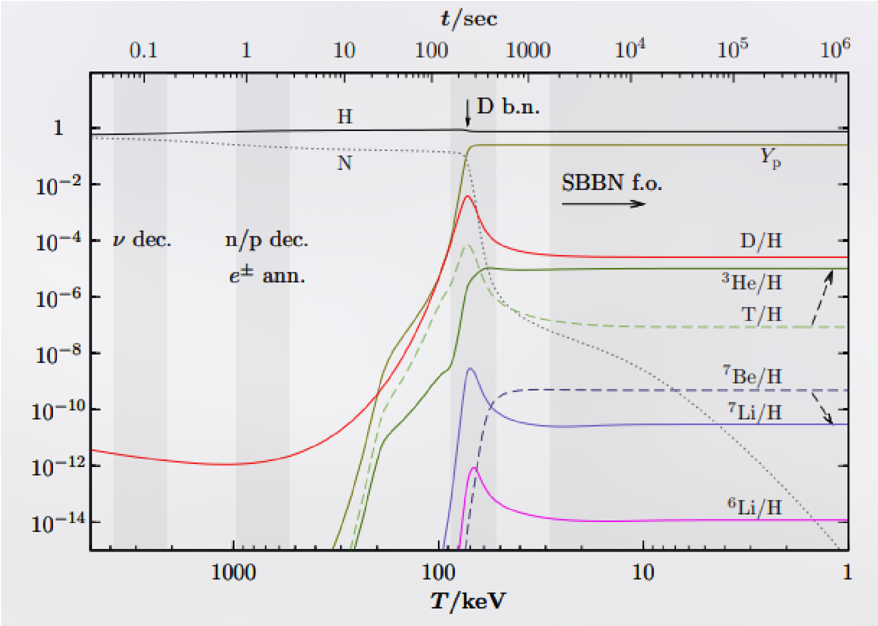

Figure 1.6: Illustration of the Big Bang Nucleosynthesis process, showing the abundances of elements with time (top-axis) and energy (bottom-axis). Taken from [45].

1.2.3 Baryon Acoustic Oscillations

During the radiation-dominated era, photons were tightly coupled to baryons through Thomson scattering of free electrons [42], forming a single photon-baryon fluid. Due to the primordial density fluctuations generated during inflation (see Section 1.1.6), over dense regions acted as gravitational potential wells, attracting baryonic matter. However, photon pressure counteracted gravitational fall. The competition between gravity and radiation pressure led to acoustic oscillations in the coupled photon-baryon fluid, generating spherical sound waves propagating outward from initial under and over densities. The speed of sound in this fluid is given by $c_s = \sqrt{\partial P/\partial \rho}$. Since the photon-baryon fluid is dominated by radiation, its equation of state is well approximated by that of a pure radiation gas, $P = \rho c^2/3$ [42], so that $\partial P/\partial \rho = c^2/3$ and $c_s \sim c/\sqrt{3}$. These oscillations ceased at recombination (see Section 1.2.4), when photons decoupled from baryons and began to free-stream. The maximum distance travelled by these sound waves, moving at $c_s \sim c/\sqrt{3}$, defines a characteristic scale, known as the sound horizon. As the CMB was emitted at $t_{CMB} \sim 380000$ years, corresponding to a redshift $1 + z \sim 1100$, the characteristic scale of the oscillations can be computed by: $c/\sqrt{3} \times t_{CMB} \times z \sim 75$ Mpc. Unfortunately, this back of the envelope computation does not include expansion during the sound wave propagation, and integrated over the emission of CMB, as decoupling between photons and baryons does not happen instantaneously, but took about 115000 years [18]. Adding these effects, the characteristic scale of BAO is about 150 Mpc. This scale is imprinted in the late-time matter distribution as a preferred clustering scale. Baryon Acoustic Oscillations (BAO) were first detected in the large-scale distribution of galaxies independently by the Sloan Digital Sky Survey [46] and the 2dF Galaxy Redshift Survey [47]. They appear as a distinct peak in the two-point correlation function at a scale of approximately 150 Mpc, corresponding to the sound horizon at recombination (see Fig. 1.7).

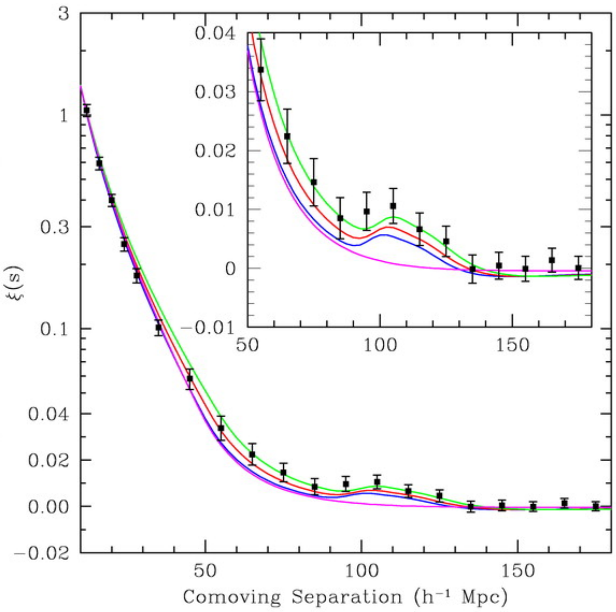

Figure 1.7: Matter two-point correlation function over comoving distance of the SDSS LRG sample. The BAO peak is visible at $110h^{-1}Mpc$, corresponding to an over density of matter at this distance. From [46].

Additionally, dark matter behaved differently, as it does not interact with photons. Consequently, dark matter is affected only by gravity, creating over densities that will be the foundation of Large Scale Structures 1.2.5.

1.2.4 Recombination

Recombination refers to the epoch when the Universe became sufficiently cool and dilute that photons could no longer efficiently maintain the coupling between matter and radiation, leading to the decoupling of the primordial plasma and the transparency of the Universe. From this moment on, photons propagate freely through space, forming a relic radiation known as the *Cosmic Microwave Background* (CMB), discussed in detail in Section 1.3. During this transition, nuclei were no longer kept ionised by high-energy photons, allowing the formation of the first neutral atoms, mainly hydrogen and helium (see Section 1.2.2).

Prior to recombination, matter and radiation were tightly coupled, and the primordial plasma was in thermal equilibrium. As a consequence, the CMB is expected to exhibit an almost perfect blackbody spectrum, described by Planck's law [48]:

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}. \tag{1.7}$$

This prediction has been confirmed by the COBE satellite, which provided the most precise measurement of a blackbody spectrum to date, as shown in Figure 1.8.

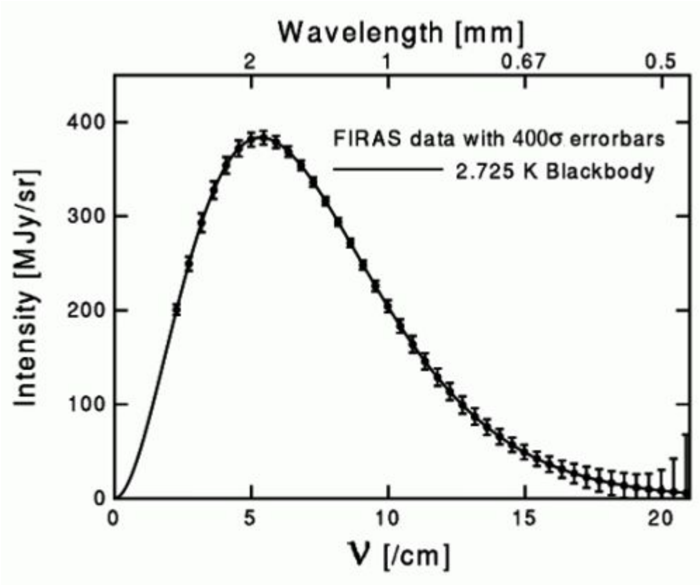

Figure 1.8: CMB spectrum measured by FIRAS instrument on COBE satellite. Plain line is the theoretical spectrum for a blackbody at 2.725 K, points are data from FIRAS with error bars multiplied by 400 to be visible.

1.2.5 Large Scale Structure

After matter–radiation decoupling, baryonic matter evolves primarily under the influence of gravity. The small primordial density perturbations grow through gravitational instability. In the linear regime, their growth is proportional to the scale factor during matter domination. Dark matter, having decoupled earlier, forms gravitational potential wells into which baryons subsequently fall, leading to the formation of structures. This hierarchical process results in the emergence of the large-scale structure of the Universe, including halos, galaxies, clusters, and filaments. On large scales, the statistical properties of matter fluctuations are described by the matter power spectrum $P(k)$, which encodes both the primordial fluctuations and their subsequent evolution through cosmic expansion and gravitational collapse. A key observable imprint of these processes is given by Baryon Acoustic Oscillations (BAO), which provide a standard ruler in the distribution of matter (see Section 1.2.3), as well as the best argument for non-baryonic dark matter.

1.3 Cosmic Microwave Background

So far, we have repeatedly referred to the Cosmic Microwave Background (see 1.2.4). However, as the CMB constitutes our primary observational probe of the primordial Universe and lies at the core of this thesis, we now present a more detailed discussion of its physical origin, its specific properties, and its connection to cosmological parameters, both through its intensity and polarisation signals.

1.3.1 Angular Power Spectrum

Before discussing the physical properties of the CMB, it is useful to introduce the mathematical framework used to study the statistical properties of the anisotropy distribution across the sky, namely the angular power spectrum. The angular power spectrum corresponds to a decomposition of a sky map into spherical harmonics and describes the variance of the anisotropies as a function of angular scale on the celestial sphere. It is equivalent to the two-point correlation function in real space. In the case of a Gaussian random field, the angular power spectrum completely characterises the statistical properties of the anisotropies, as higher-order correlations are completely determined by 2nd order through Wick's theorem [49, 42].

Spherical Harmonics

Spherical harmonics are the eigenfunctions of the angular part of the Laplace operator on the sphere:

$$[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}]Y_{\ell m}(\theta,\phi) = -\ell(\ell+1)Y_{\ell m}(\theta,\phi). \tag{1.8}$$

They form a complete orthonormal basis on the sphere, which makes them particularly well suited to decompose any signal defined on the sky, such as CMB temperature anisotropies. The spherical harmonic functions $Y_{\ell m}(\theta,\phi)$ depend on the angular coordinates θ and ϕ, as well as on two integers: the multipole $\ell \in \mathbb{N}$, which characterises the angular scale on the sphere, and the azimuthal number m, such that $-\ell \leq m \leq \ell$. For each multipole ℓ, there are $2\ell+1$ independent modes. The angular scale associated with a given multipole is roughly given by $\theta \sim \pi/\ell$. The spherical harmonics can be written explicitly as:

$$Y_{\ell m}(\theta,\phi) = \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}P_\ell^m(\cos\theta)e^{im\phi}, \tag{1.9}$$

where P_ℓ^m are the associated Legendre polynomials (see e.g. [50] for more details).

Angular Power Spectrum

After introducing spherical harmonics, the temperature anisotropies (or any scalar field on the sphere) can be expanded as:

$$\Delta T(\vec{n}) = \sum_{\ell=0}^{+\infty}\sum_{m=-\ell}^{\ell} a_{\ell m}Y_{\ell m}(\vec{n}). \tag{1.10}$$

Using the orthogonality of spherical harmonics, the coefficients $a_{\ell m}$ are given by:

$$a_{\ell m} = \int_{4\pi} \Delta T(\vec{n})Y^*_{\ell m}(\vec{n})d\Omega \tag{1.11}$$

$$\approx \Omega_{\rm pix}\sum_{p=1}^{N_{\rm pix}} \Delta T(p)Y^*_{\ell m}(p), \tag{1.12}$$

where the second line corresponds to a uniformly discretised sky map, with Ω_{pix} the size of one pixel.

The statistical properties of the field are fully described by the two-point correlation function for a Gaussian field. In harmonic space, this implies:

$$\langle a^*_{\ell m} a_{\ell' m'} \rangle = \delta_{\ell\ell'} \delta_{mm'} C_\ell, \tag{1.13}$$

with $\langle\rangle$ designating ensemble-average over many sky realisations, and C_ℓ is the angular power spectrum.

For a statistically isotropic Gaussian random field, the harmonic coefficients $a_{\ell m}$ are independent Gaussian random variables with zero mean and variance C_ℓ. An unbiased estimator of the angular power spectrum is then given by:

$$\hat{C}_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2, \tag{1.14}$$

which satisfies:

$$\langle \hat{C}_\ell \rangle = C_\ell. \tag{1.15}$$

Additionally, we are used to expressing C_ℓ in a rescaled form. Indeed, as primordial fluctuations are expected to be scale-invariant, one can show that $C_\ell \propto \frac{1}{\ell(\ell+1)}$. This motivates the definition of the scale-invariant power spectrum $D\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell$, which provides a more intuitive representation of the angular power spectrum.

Cosmic variance

Unfortunately, we only have access to a single realisation of the sky, and independent modes for each multipole ℓ. Meaning that there is an intrinsic statistical uncertainty on the estimator $\hat{C}_\ell$, which cannot be reduced by improving instrumental sensitivity. This fundamental limitation is known as *cosmic variance*. The variance of the estimator $\hat{C}_\ell$ can be computed as:

$$\begin{aligned}
\frac{\mathrm{Var}(\hat{C}_\ell)}{C_\ell^2} &= \frac{\langle (\hat{C}_\ell - C_\ell)^2 \rangle}{C_\ell^2} \\
&= \frac{1}{(2\ell+1)^2 C_\ell^2} \left\langle \sum_{mm'} a^*_{\ell m} a_{\ell m}\, a^*_{\ell m'} a_{\ell m'} \right\rangle - 1
\end{aligned} \tag{1.16}$$

From Wick's theorem, the four-point function of a zero-mean Gaussian field factorises into products of two-point functions:

$$\left\langle \sum_{mm'} a^*_{\ell m} a_{\ell m}\, a^*_{\ell m'} a_{\ell m'} \right\rangle = \sum_{mm'} \Big(\langle a^*_{\ell m} a_{\ell m}\rangle \langle a^*_{\ell m'} a_{\ell m'}\rangle + \langle a^*_{\ell m} a_{\ell m'}\rangle \langle a_{\ell m} a^*_{\ell m'}\rangle + \langle a^*_{\ell m} a^*_{\ell m'}\rangle \langle a_{\ell m} a_{\ell m'}\rangle \Big)$$
$$= C_\ell^2 \sum_{mm'} \left(1 + \delta_{mm'} + \delta_{m,-m'}\right)$$
$$= C_\ell^2 \left[(2\ell+1)^2 + 2(2\ell+1)\right], \tag{1.17}$$

where the last two terms use $\langle a^*_{\ell m} a_{\ell m'}\rangle = C_\ell\, \delta_{mm'}$ and the condition $a^*_{\ell m} = (-1)^m a_{\ell,-m}$, which gives $\langle a^*_{\ell m} a^*_{\ell m'}\rangle \langle a_{\ell m} a_{\ell m'}\rangle = C_\ell^2\, \delta_{m,-m'}$. Substituting back:

$$\frac{\mathrm{Var}(\hat{C}_\ell)}{C_\ell^2} = \frac{C_\ell^2 \left[(2\ell+1)^2 + 2(2\ell+1)\right]}{(2\ell+1)^2 C_\ell^2} - 1$$
$$= \frac{2\ell+3}{2\ell+1} - 1 = \frac{2}{2\ell+1}.$$

Then, cosmic variance can be computed using:

$$\frac{\mathrm{Var}(\hat{C}_\ell)}{C_\ell^2} = \frac{2}{2\ell+1}. \tag{1.18}$$

Sky pixelisation

In real observations, sky maps must be pixelised to allow for numerical treatment; consequently, the angular coordinates θ and ϕ are discretised. The most widely used pixelisation scheme in CMB experiments (and cosmology in general) is the HEALPix method [51], which is accessible in the Python package Healpy[2]. The number of pixels in HEALPix, and thus the angular resolution, is controlled by the parameter N_{side}, with: $N_{\mathrm{pix}} = 12 N_{\mathrm{side}}^2$. In principle, the maximum multipole that can be reliably probed depends on the pixel size. In practice, HEALPix relies on numerical approximations to accelerate spherical harmonic transforms, which become computationally expensive at high multipoles. To ensure accuracy, the maximum multipole is typically limited to $\ell_{\mathrm{max}} = 3N_{\mathrm{side}}$.

This framework is based on three key properties:

1. **Hierarchical structure.** Pixels are organized in a tree structure: increasing the resolution corresponds to subdividing each pixel into four sub-pixels. This allows for efficient multi-resolution analysis and facilitates fast local operations such as convolutions. Moreover, the indexing scheme preserves locality, meaning that nearby points on the sphere have nearby indices, which enables efficient neighbour searches.

2. **Equal area.** All pixels have exactly the same area, $\Omega_{\mathrm{pix}} = \frac{4\pi}{N_{\mathrm{pix}}}$, which ensures that white noise remains white in pixel space and avoids introducing spatial biases in the estimation of C_ℓ. It also simplifies numerical integrations on the sphere: $\int f(\hat{n})\, d\Omega \approx \Omega_{\mathrm{pix}} \sum_p f_p$.

[2] https://github.com/healpy/healpy

3. **Iso-latitude.** The centres of the pixels are arranged on rings of constant latitude, each containing pixels uniformly distributed in ϕ. This property significantly accelerates the computation of the spherical harmonic coefficients $a_{\ell m}$, since the associated Legendre polynomials $P_\ell^m(\cos\theta)$ only need to be evaluated once per ring. As a result, the computational complexity of spherical harmonic transforms is reduced from $\mathcal{O}(N_{\rm pix}\ell_{\rm max}^2)$ to $\mathcal{O}(N_{\rm pix}^{1/2}\ell_{\rm max}^2)$.

Beyond pixelisation, HEALPix provides a comprehensive set of tools for spherical data analysis, including efficient algorithms for spherical harmonic transforms and angular power spectrum estimation, which are widely used in CMB studies and throughout this thesis.

1.3.2 CMB Temperature

As discussed above (see 1.2.4), the CMB follows a nearly perfect blackbody spectrum. Its emission is therefore entirely determined by a single parameter: its temperature. Since the temperature of the photon–baryon plasma at recombination depends on local physical conditions, the observed CMB temperature anisotropies directly trace spatial fluctuations in this plasma. For this reason, CMB intensity anisotropies are conventionally expressed in terms of an effective temperature, which encodes information about the density fluctuations of the primordial plasma and, more generally, about its underlying physics.

Temperature Monopole and Dipole

The temperature monopole of the CMB is its mean temperature across the sky. It has been accurately measured by FIRAS instrument of COBE satellite at $T_0 = 2.7255 \pm 0.0006K$ [31]. This measurement remains the most precise determination of the CMB monopole, as subsequent experiments have primarily focused on measuring anisotropies around this mean temperature, with the notable exception of dedicated spectral distortion[3] missions (PIXIE [52], PRISTINE, FOSSIL [53]), aiming to improve constraints on the monopole spectrum itself. From particle physics considerations, photon–baryon decoupling occurs when the primordial plasma reaches a temperature $T_{\rm rec} \sim 3000K$. As discussed in 1.1.2, cosmic expansion induces a redshift of photon wavelengths. For a blackbody spectrum, this implies that the temperature scales with redshift as:

$$T(z) = T_0(1+z), \tag{1.19}$$

where T_0 is the present CMB temperature and z its redshift. Using the measured value of T_0 and the computed recombination temperature $T_{\rm rec}$, one can infer that the CMB was emitted at a redshift $z_{\rm rec} \sim 1100$.

A first level of anisotropy is induced by the motion of the observer with respect to the CMB rest frame. Because the Solar System has a peculiar velocity relative to this frame, arising from its motion within the Milky Way and from the motion of the

[3] Spectral distortions of the CMB spectrum occur when energy is injected into the photon bath at epochs when thermalisation processes become inefficient. The CMB spectrum then no longer follows a blackbody law but different distributions depending on the epoch of injection.

Milky Way within the Local Group, the observed CMB temperature is Doppler shifted across the sky [6]. This effect produces a dipolar anisotropy, known as the CMB solar dipole. The dipole anisotropy was first detected in 1977 [54], then confirmed by the DMR instrument aboard the COBE satellite [55], and later measured with greater precision by WMAP [56] and Planck Collaboration [57]. The solar dipole, simulated using PySM, is shown in Figure 1.9.

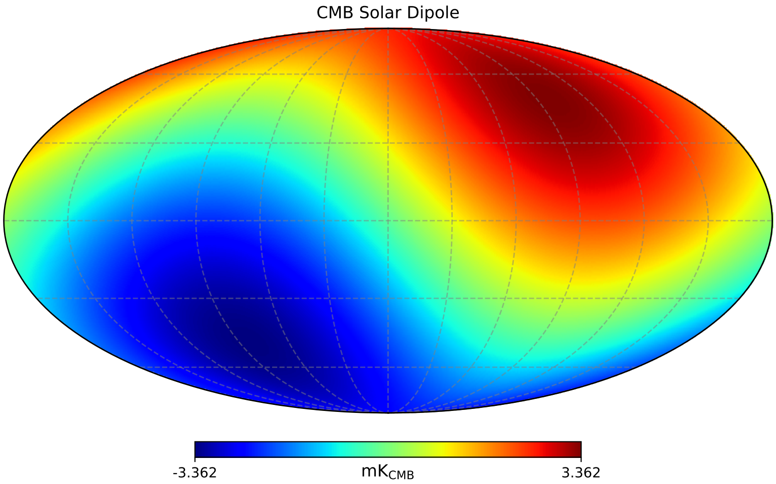

Figure 1.9: CMB solar dipole simulated using the PySM software [58] and calibrated using Planck 2018 measurements [57].

CMB Anisotropies

As the primordial plasma exhibits small density inhomogeneities at the level of $\sim 10^{-5}$, these are expected to manifest as temperature anisotropies in the Cosmic Microwave Background (CMB). The first detection of these anisotropies was made by the COBE satellite using its DMR instrument [55]. The full-sky maps obtained after four years of observation are shown in Figure 1.10.

These measurements, with a 7° angular resolution at 90 GHz, were significantly improved by later experiments, in particular the Planck satellite, which achieved higher sensitivity and angular resolution in CMB temperature maps (0.16° at 100 GHz, 0.12° at 150 GHz). The 150 GHz full-sky map is shown in Figure 1.11, taken from the Planck Legacy Archive[4].

The Planck satellite reached a regime in which the measurement of the temperature power spectrum is cosmic-variance limited over a wide range of angular scales. In this regime, the dominant source of uncertainty is no longer instrumental noise, but the fundamental statistical variance associated with observing a single realisation of

[4] `https://pla.esac.esa.int/`

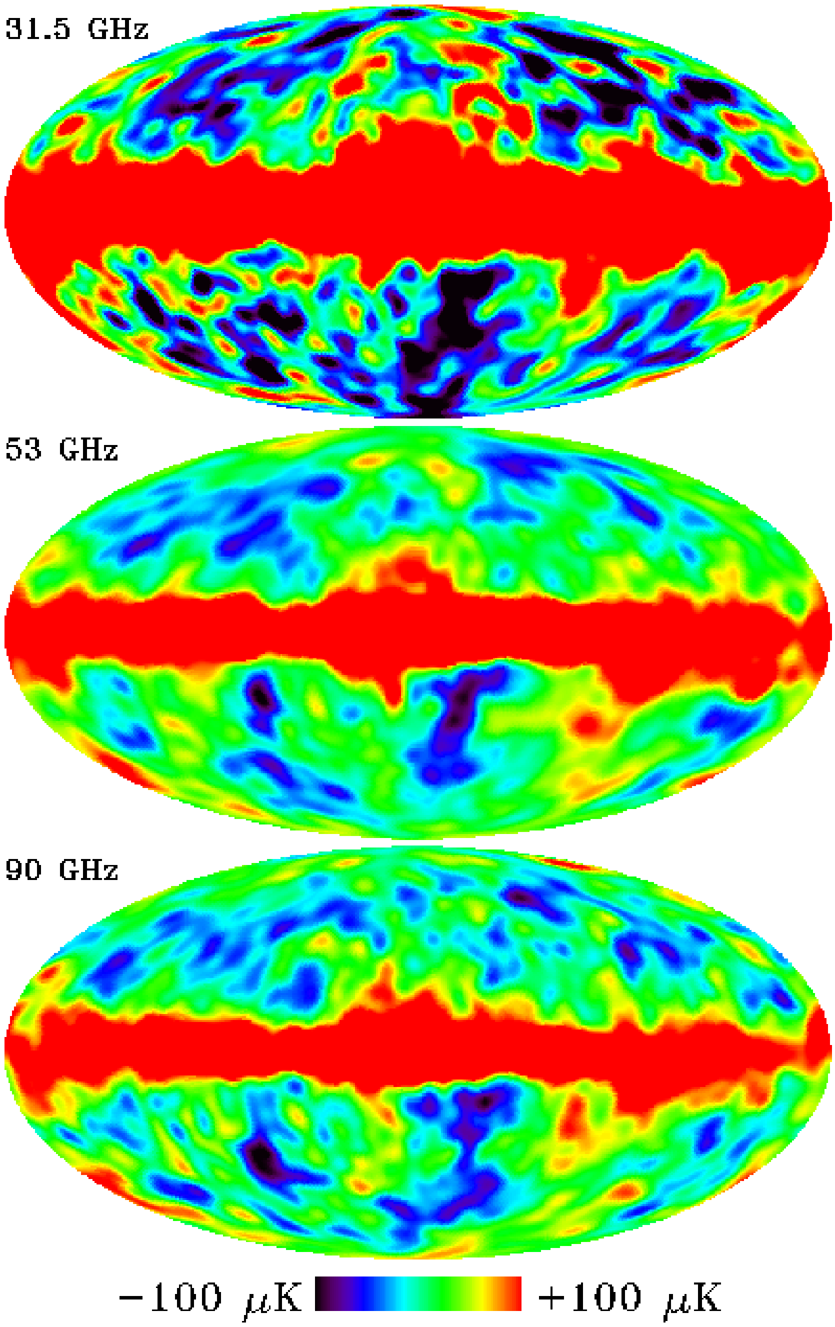

Figure 1.10: Full-sky maps from the DMR instrument aboard the COBE satellite after four years of observation at 31.5, 53 and 90 GHz. The monopole and dipole components have been removed. The bright band along the Galactic plane is due to Galactic emission, while cosmological temperature anisotropies are visible at high latitudes. Maps taken from [59].

the Universe. However, this limitation does not apply uniformly: at high multipoles, instrumental noise and foreground contamination become significant, while in polarisation, especially B-modes, cosmic-variance-limited performance has not been reached.

The new generation of ground-based CMB experiments aims to improve measurements at small angular scales (high multipoles), as discussed in Section 1.5.

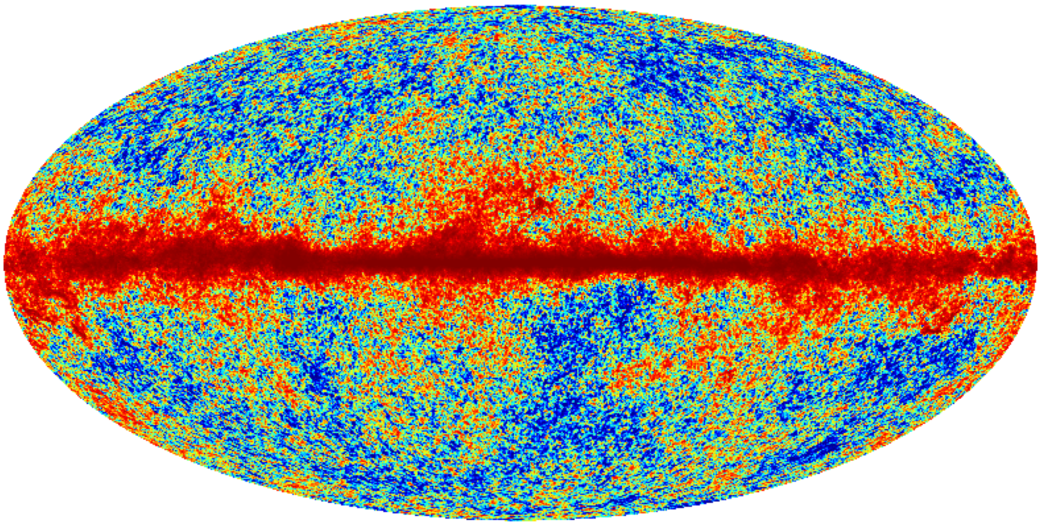

Figure 1.11: Full-sky CMB temperature map from the Planck satellite at 150 GHz. The monopole and dipole components have been removed. Compared to COBE, the improvement in angular resolution is clearly visible.

CMB anisotropy measurements are a key probe to understand the physics of the primordial plasma and, more generally, the mechanisms of the early Universe.

CMB power spectrum

The CMB anisotropies are expected to follow nearly Gaussian statistics, as expected by Inflation [60]. Observations by the Planck satellite have found no significant evidence for primordial non-Gaussianity in the temperature fluctuations. This justifies the use of the angular power spectrum as a complete statistical descriptor of the CMB anisotropies. From the Planck maps, the angular power spectrum shown in Figure 1.12 can be reconstructed. The large uncertainties at low multipoles (ℓ), corresponding to large angular scales, arise primarily from cosmic variance, as discussed previously. They are further exacerbated by observational limitations: emission from the Milky Way obscures part of the sky, preventing full-sky measurements and limiting access to the lowest multipoles of the CMB power spectrum. One can note that y-axis is showing D_ℓ instead of C_ℓ, D_ℓ only being a re-scaling of the angular power spectrum, as it represents the contribution to the variance per logarithmic interval in multipole ℓ. In particular, a scale-invariant primordial spectrum, as expected from Inflation (Harrisson-Zel'dovitch spectrum [61, 62]) corresponds to a nearly constant D_ℓ at large angular scales, which is given by:

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell. \tag{1.20}$$

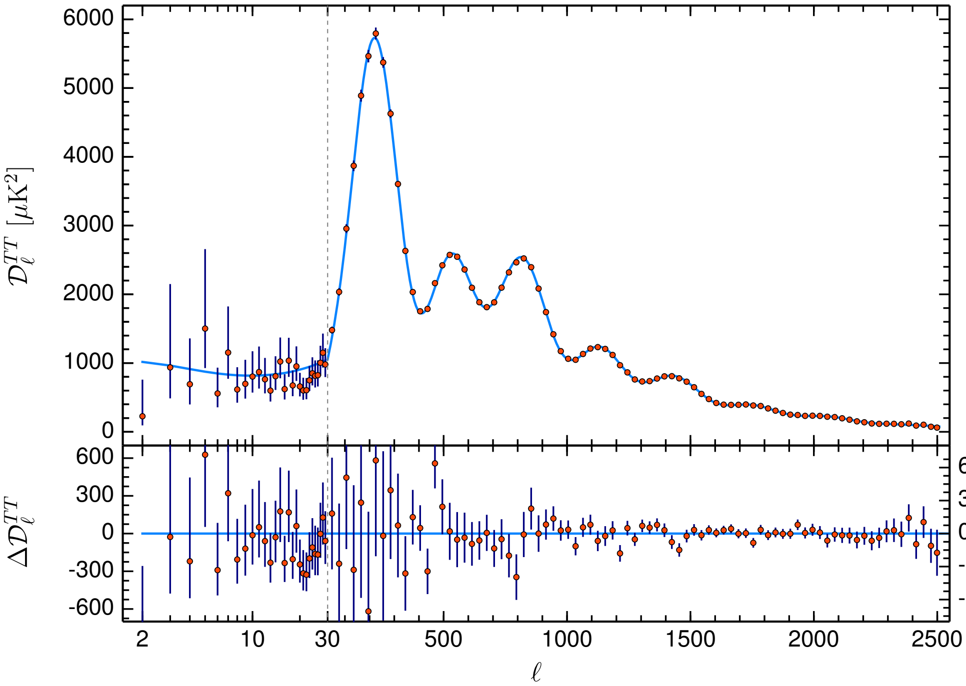

Figure 1.12: (Top) Best fit of the Angular power spectrum from Planck Temperature maps according to the minimal ΛCDM model. (Bottom) shows the difference between power spectrum computed from data and the best fit. Taken from [18].

The CMB angular power spectrum exhibits three main regimes, which we describe below.

Sachs-Wolfe plateau: $\ell \lesssim 30$. At angular scales larger than the first acoustic peak, perturbation modes were outside the sound horizon at recombination and could not undergo causal acoustic oscillations. Temperature anisotropies therefore directly reflect the primordial matter power spectrum, unprocessed by BAO physics. The Harrison-Zel'dovich spectrum [61, 62] predicts a scale-invariant primordial spectrum ($n_s = 1$), which via the Sachs-Wolfe effect [63] ($\delta T/T \approx \Phi/3$ for adiabatic perturbations) produces the nearly flat plateau in D_ℓ. Measurements in this regime are limited by cosmic variance and partial sky coverage.

Baryon Acoustic Oscillations: $30 \lesssim \ell \lesssim 2000$. The peaks in the CMB power spectrum arise from acoustic oscillations (see 1.2.3) in the tightly coupled photon-baryon plasma prior to recombination. The first peak, at $\ell \sim 220$, corresponds to the mode that has undergone half an oscillation at the time of last scattering. Subsequent peaks correspond to higher-order oscillations: odd peaks correspond to compressions in gravitational potentials, while even peaks correspond to rarefactions driven by photon pressure.

Silk Damping: $\ell \gtrsim 1000-2000$. At small angular scales, anisotropies are suppressed due to photon diffusion (Silk damping) and the finite thickness of the last scattering surface. Photons undergo a random walk before decoupling, which erases small-scale temperature fluctuations. This leads to an exponential suppression of power at high multipoles [64].

1.3.3 CMB Polarisation

So far, we discussed the temperature anisotropies of the Cosmic Microwave Background. In addition, primordial mechanisms are also polarising CMB signal, up to 10% of the temperature amplitude signal. Studying the polarisation part of the signal provide rich information on primordial Universe physics. We describe here these associated mechanisms and their implications on CMB signal.

Reminder on polarisation

Polarisation of the CMB plays a central role in this thesis. We briefly introduce here the relevant definitions and properties used in the following. Light, including the CMB, is an electromagnetic wave, consisting of electric and magnetic fields governed by Maxwell's equations [65]. In Cartesian coordinates (x, y, z), an electric field propagating along the z-axis can be written as:

$$\vec{E}(t) = \begin{pmatrix} A_x e^{i(\omega t+\phi_x)} \\ A_y e^{i(\omega t+\phi_y)} \\ 0 \end{pmatrix}, \tag{1.21}$$

where A_x and A_y are the amplitudes, ω is the frequency, and ϕ_x, ϕ_y are the phases.

Stokes parameters The polarisation state of the radiation is fully described by the Stokes parameters [66], which can be defined through the coherence matrix:

$$C = \begin{pmatrix} \langle |E_x|^2 \rangle & \langle E_x E_y^* \rangle \\ \langle E_x^* E_y \rangle & \langle |E_y|^2 \rangle \end{pmatrix} \tag{1.22}$$

$$= \frac{1}{2} \begin{pmatrix} I+Q & U+iV \\ U-iV & I-Q \end{pmatrix}. \tag{1.23}$$

From this definition, the Stokes parameters are given by:

$$I = \langle |E_x|^2 \rangle + \langle |E_y|^2 \rangle, \tag{1.24}$$

$$Q = \langle |E_x|^2 \rangle - \langle |E_y|^2 \rangle, \tag{1.25}$$

$$U = 2\,\mathrm{Re}\left(\langle E_x E_y^* \rangle\right), \tag{1.26}$$

$$V = 2\,\mathrm{Im}\left(\langle E_x E_y^* \rangle\right). \tag{1.27}$$

The parameter I corresponds to the total intensity of the radiation. The parameters Q and U describe linear polarisation, while V characterises circular polarisation. In the context of the CMB, no significant circular polarisation has been detected, nor theoretically predicted from Standard Physics, and V is therefore neglected.

A visual representation of the Stokes parameters is shown in Figure 1.13.

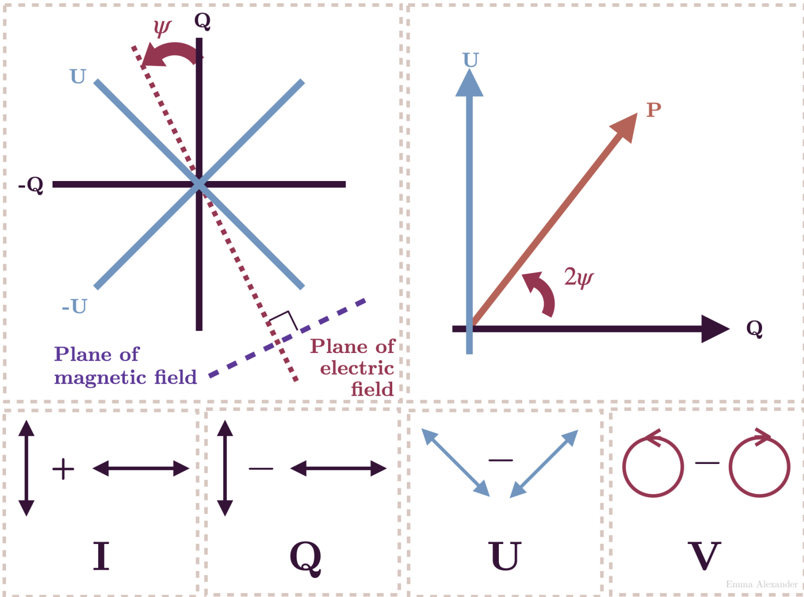

Figure 1.13: Visual representation of the Stokes parameters. Image from Wikimedia Commons [67].

E- and B-modes The Stokes parameters Q and U depend on the choice of local coordinate system and are therefore not invariant under rotations on the sphere. Under a rotation by an angle ψ, they transform as:

$$(Q \pm iU)'(\hat{n}) = e^{\mp 2i\psi}(Q \pm iU)(\hat{n}), \tag{1.28}$$

which shows that they are spin-2 quantities.

To construct rotationally invariant quantities, [68] introduced a decomposition in terms of spin-weighted spherical harmonics:

$$(Q \pm iU)(\hat{n}) = \sum_{\ell m} a_{\ell m}^{\pm 2} \, {}_{\pm 2}Y_{\ell m}(\hat{n}). \tag{1.29}$$

From these coefficients, the scalar E-mode and pseudo-scalar B-mode components are defined as:

$$a_{\ell m}^{E} = -\frac{1}{2}\left(a_{\ell m}^{2} + a_{\ell m}^{-2}\right), \tag{1.30}$$

$$a_{\ell m}^{B} = \frac{i}{2}\left(a_{\ell m}^{2} - a_{\ell m}^{-2}\right). \tag{1.31}$$

The corresponding real-space fields are then:

$$E(\hat{n}) = \sum_{\ell m} a^E_{\ell m} Y_{\ell m}(\hat{n}), \tag{1.32}$$

$$B(\hat{n}) = \sum_{\ell m} a^B_{\ell m} Y_{\ell m}(\hat{n}). \tag{1.33}$$

E-modes correspond to gradient-like patterns and are even under parity transformations. They are primarily generated by scalar perturbations. In contrast, B-modes corresponding to curl-like patterns, are odd under parity, and can be generated by tensor perturbations (primordial gravitational waves) as well as by gravitational lensing of the E-modes[5].

A visual representation of the E- and B-mode patterns is shown in Figure 1.14.

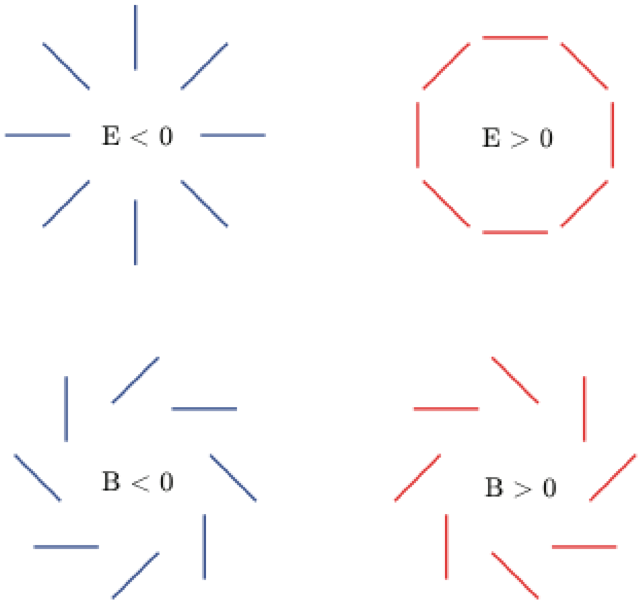

Figure 1.14: Visual representation of the E and B modes. Image from Wikimedia Commons [69].

Compton Scattering

Compton scattering, discovered in 1923 by Arthur H. Compton [70], describes the interaction between a photon and a charged particle, typically an electron. In the context of the CMB, the relevant regime is the low-energy limit of this interaction, known as Thomson scattering, in which photons scatter elastically off free electrons.

In this regime, the scattering cross-section depends on the angle between the incoming and outgoing photon directions, and on the polarisation state of the radiation. In particular, the differential cross-section follows

$$\frac{d\sigma}{d\Omega} \propto |\boldsymbol{\epsilon}' \cdot \boldsymbol{\epsilon}|^2, \tag{1.34}$$

[5] Their name was given by analogy to the electric and magnetic fields' properties.

where $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'$ are the incoming and outgoing polarisation vectors. This angular dependence implies that radiation scattered at 90° is linearly polarised.

In the early Universe, prior to recombination, photons and baryons form a tightly coupled plasma, with photons scattering frequently off free electrons. Each electron receives radiation from all directions. If the incoming radiation field is perfectly isotropic (monopole), or possesses only a dipole anisotropy, the symmetry of the configuration ensures that no net polarisation is produced after scattering.

However, if the radiation field exhibits a quadrupole anisotropy, corresponding to a four-lobed angular pattern, this symmetry is broken. In this case, the scattered radiation has different intensities along orthogonal directions, leading to a net linear polarisation. Therefore, Thomson scattering generates polarisation only in the presence of a non-zero quadrupole moment in the radiation field.

At the surface of last scattering, local temperature anisotropies naturally generate such quadrupole patterns from tidal effect of e^- falling into potential wells, resulting in the production of CMB polarisation, see Figure 1.15. This mechanism produces exclusively E-mode polarisation; B-modes are not generated directly by Thomson scattering, but arise from primordial tensor perturbations or secondary effects such as gravitational lensing.

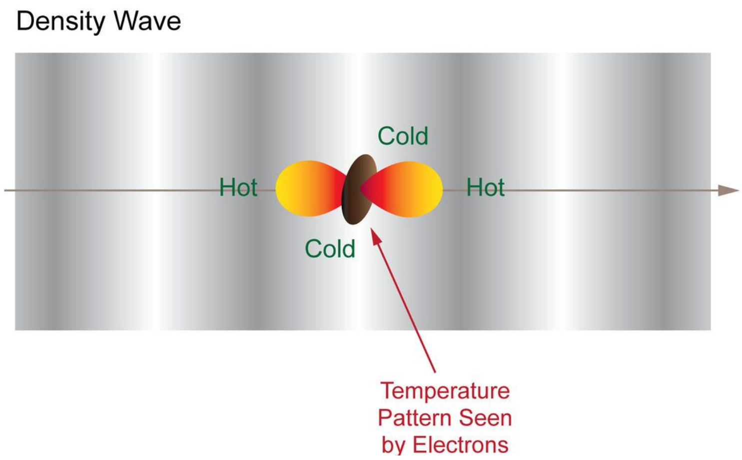

Figure 1.15: Density perturbations induce a local quadrupole anisotropy in the radiation field seen by an electron. Figure from BICEP2 Collaboration.

Compton scattering, along with monopole, dipole, and quadrupole configurations, are illustrated in Figure 1.16.

Gravitational Waves

Primordial gravitational waves, predicted in the context of inflationary cosmology, provide an additional source of anisotropy in the radiation field at the time of last scattering. These tensor perturbations correspond to transverse, traceless distortions of spacetime that propagate as waves, periodically stretching and compressing space along orthogonal directions. As gravitational waves propagate through the primordial plasma, they induce a time-dependent quadrupole anisotropy in the photon distribution seen by free electrons. Through Thomson scattering, this quadrupole pattern generates

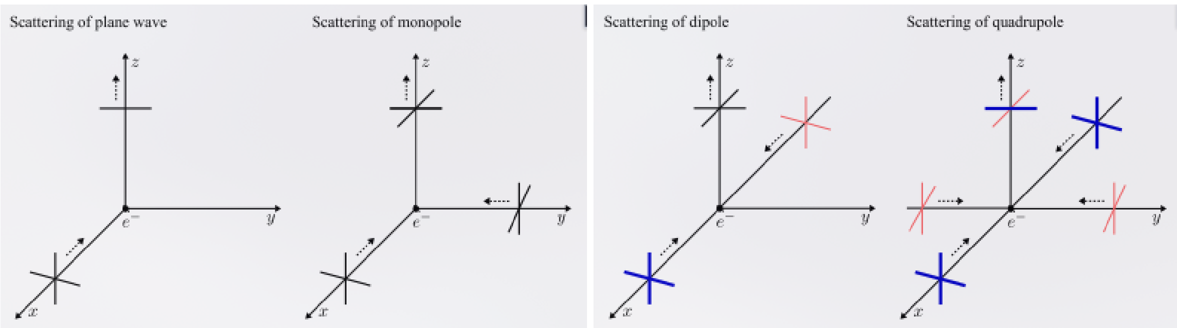

Figure 1.16: (left) Compton scattering of photon coming from x-axis, scattered by an electron in z direction. Only y component remains, leading to linear polarisation. (left centre) Monopole configuration. Isotropic radiation produces no polarisation. (right centre) Dipole configuration. Blue emphasise hotter photon and red colder photon. The resulting radiation after scattering travelling along z-axis in black is the average of x-axis radiations. (right) Quadrupole configuration. Scattered radiation has higher intensity along x-axis than along y-axis, leading to linear polarisation. Taken from [42].

linear polarisation in the same way as for scalar perturbations. However, unlike density (scalar) perturbations, tensor perturbations possess a handedness associated with their two polarisation states, which introduces a curl component in the resulting polarisation field. As a consequence, primordial gravitational waves generate both E-mode and B-mode polarisation. In particular, they are the only known mechanism capable of producing primordial B-modes at large angular scales. The amplitude of this signal is directly related to the energy scale of inflation and is commonly parametrised by the tensor-to-scalar ratio $r = \frac{P_{tensor}(k_0)}{P_{scalar}(k_0)}$, defined as the ratio between amplitude of tensorial and scalar perturbations. Detecting this B-mode component is therefore a major objective of modern CMB experiments, as it would provide direct evidence for inflation and insight into physics at very high energies.

1.3.4 Cosmological parameters

The minimal ΛCDM model is described by six free cosmological parameters. Although the model can be extended to include additional parameters, such extensions often introduce degeneracies and are not required to provide a significantly better fit to CMB observations [18]. The six parameters of the ΛCDM model are:

- **Baryon density** $\Omega_b h^2$,
- **Cold dark matter density** $\Omega_c h^2$,
- **Angular size of the sound horizon at recombination** θ_{MC},
- **Optical depth to reionization**[6] τ,
- **Amplitude of scalar fluctuations** $\ln(10^{10} A_s)$,

[6]Reionization occurred after recombination, when the first luminous sources formed and emitted high-energy photons that progressively ionised the neutral hydrogen in the intergalactic medium.

- **Spectral index of scalar fluctuations n_s.**

The parameters $\Omega_b h^2$ and $\Omega_c h^2$ represent the physical baryon and cold dark matter densities. Since the critical density scales as $\rho_c \propto H_0^2$, the actual matter densities depend on the combination ΩH_0^2. Using $h = H_0/(100\ kms^{-1}Mpc^{-1})$ therefore leads naturally to the dimensionless quantities Ωh^2, which are directly constrained by CMB.

In order to constrain these parameters, additional assumptions are typically made. The number of relativistic neutrino species is fixed to $N_{\rm eff} = 3.046$ [71], and the sum of neutrino masses is assumed to be $\sum m_\nu = 0.06\,\mathrm{eV}$ [72]. The Universe is assumed to be spatially flat ($\Omega_k = 0$), and dark energy is modelled as a cosmological constant ($w = -1$). The resulting constraints on these parameters from the Planck 2018 data, combining temperature and polarisation power spectra, are shown in Figure 1.17 and in Table 1.1. From these fitted values, other cosmological parameters can be derived, such as the total matter density Ω_m, the Hubble parameter H_0, and the age of the Universe t_0.

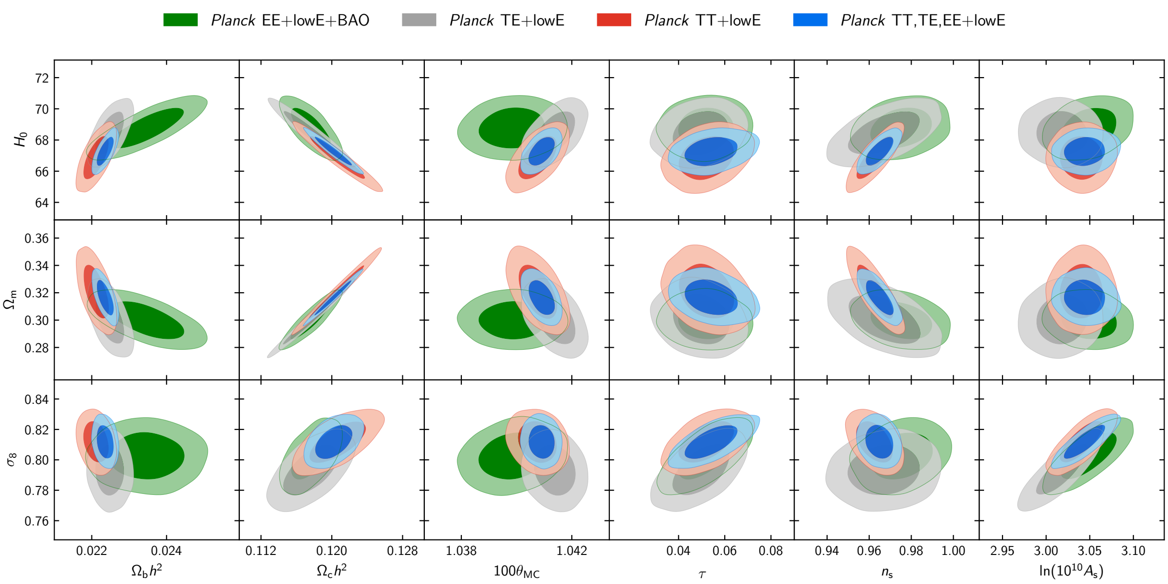

Figure 1.17: Constraints on the base parameters of the ΛCDM model using Planck 2018 data, taken from [18].

1.4 CMB contaminations

Several physical mechanisms generate B-mode polarisation on the sky. In the context of CMB observations, these additional signals act as contaminants that must be separated from the cosmological signal of interest (although they are fascinating from an astrophysical perspective). This process is known as *component separation*. The development of component separation algorithms constitutes a central part of this thesis. Before introducing these methods, it is necessary to describe the main sources of contamination and to understand their physical origin and statistical properties.

Parameter	Combined
Base parameters	
$\Omega_b h^2$	0.02233 ± 0.00015
$\Omega_c h^2$	0.1198 ± 0.0012
$100\,\theta_{\mathrm{MC}}$	1.04089 ± 0.00031
τ	0.0540 ± 0.0074
$\ln(10^{10} A_s)$	3.043 ± 0.014
n_s	0.9652 ± 0.0042
Derived parameters	
$\Omega_m h^2$	0.1428 ± 0.0011
H_0 [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	67.37 ± 0.54
Ω_m	0.3147 ± 0.0074
Age [Gyr]	13.801 ± 0.024
σ_8	0.8101 ± 0.0061
$S_8 \equiv \sigma_8\,(\Omega_m/0.3)^{0.5}$	0.830 ± 0.013
z_{re}	7.64 ± 0.74
$100\,\theta_*$	1.04108 ± 0.00031
r_{drag} [Mpc]	147.18 ± 0.29

Table 1.1: Base ΛCDM cosmological parameters from Planck 2018 temperature and polarisation power spectra, from [18]. The upper block lists the six sampled parameters; the lower block lists derived quantities. Values are marginalized means $\pm\,1\sigma$, averaged over the *Plik* and *CamSpec* likelihood analyses.

1.4.1 Lensing

As described by General Relativity [1], mass curves space-time, and this curvature acts as a gravitational lens that deflects the propagation of light. In particular, *gravitational lensing* alters the polarisation of photons. In the context of the CMB, photons travelling toward us are deflected by large-scale structures, inducing a mixing between E- and B-mode polarisation. Given the relative amplitudes of these two components, the conversion of B-modes into E-modes is negligible. In contrast, the leakage of E-modes into B-modes constitutes a major obstacle for the detection of primordial B-modes. Figure 1.18 compares the B-mode power spectrum (in blue) for different values of the tensor-to-scalar ratio r (e.g. $r = 0.01$ and $r = 0.001$) with the lensing-induced B-mode spectrum (in green). It shows that if $r \lesssim 10^{-2}$, as suggested by current constraints [73], lensing contamination outpaces a significant fraction of the primordial B-mode signal. To overcome this limitation, the lensing contribution must be removed using a technique known as delensing. This approach consists in reconstructing a template of the lensing signal from measurements of E-modes and of the projected gravitational potential, and subtracting this template from the observed B-mode map.

1.4.2 Astrophysical Dust

Thermal dust emission constitutes one of the main astrophysical sources of B-mode polarisation, in addition to gravitational lensing of the CMB. In the interstellar medium, aspherical dust grains form large-scale structures and tend to align partially with the Galactic magnetic field through radiative torque mechanisms. As a result, their thermal emission is linearly polarised. Heated by stellar radiation to typical temperatures of $T_d \sim 20K$, these grains emit in the microwave and far-infrared following, in the most

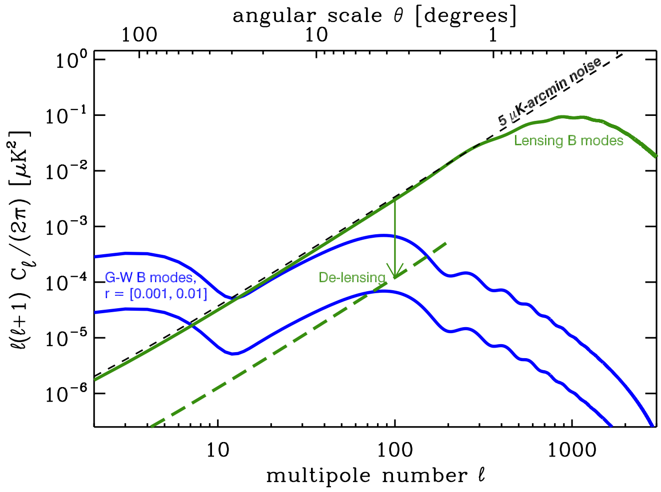

Figure 1.18: Blue curves are the power spectrum of B-modes for different value of the tensor-to-scalar ratio(0.01 and 0.001). The green curve is the power spectrum of lens-induced E-to-B mixing. Delensing can reduce the amplitude of this effect by large factors (green dashed curve) yielding lower effective noise in B-mode maps. Taken from [74].

simplistic model called d_0, a modified blackbody (grey body) law [75]. This emission can be modelled as:

$$I_{\nu,d} \propto \nu^{\beta_d} B_\nu(T_d), \tag{1.35}$$

where β_d is the dust spectral index and $B_\nu(T_d)$ is the Planck function, describing the blackbody emission. At frequencies above $\sim 100GHz$, thermal dust emission dominates over CMB polarisation, as shown in Figure 1.19. Its spectral energy distribution (SED) is often parameterised as:

$$f_d(\nu, T_d) = A_d(\frac{\nu}{\nu_{0,d}})^{\beta_d} \frac{e^{\frac{h\nu}{kT_d}} - 1}{e^{\frac{h\nu_{0,d}}{kT_d}} - 1}, \tag{1.36}$$

with A_d is the amplitude at reference frequency $\nu_{0,d}$ and β_d is the spectral index. In practice, the dust temperature is often fixed to $T_d \approx 20K$, due to degeneracies with A_d and β_d while fitting dust SED. Figures 1.20 and 1.21 shows respectively Intensity and Polarisation dust maps at $353GHz$, as measured by Planck and reconstructed using the *Commander* [7] framework [77].

1.4.3 Synchrotron

Synchrotron radiation arises from relativistic electrons spiralling in Galactic magnetic fields. As these charged particles are accelerated perpendicular to their velocity, they emit highly anisotropic radiation that is intrinsically linearly polarised. This mechanism constitutes one of the main low-frequency astrophysical foregrounds contaminating CMB polarisation measurements.

[7] `https://github.com/Cosmoglobe/Commander`

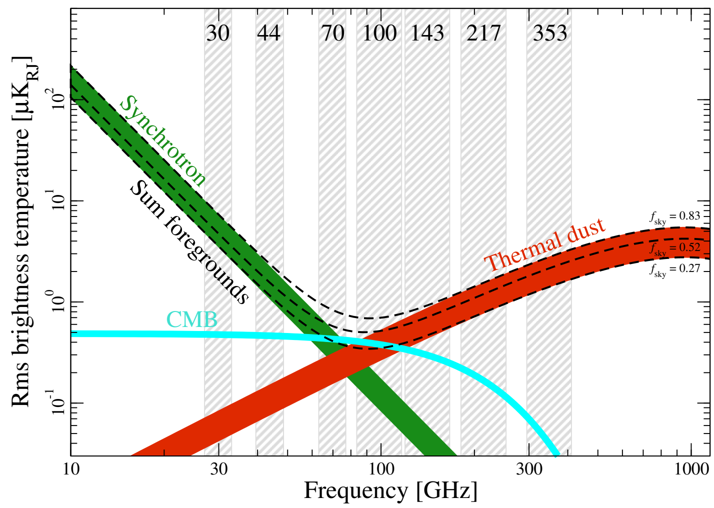

Figure 1.19: Planck's measurements of polarisation amplitude RMS as a function of frequency for polarised synchrotron emission (green), polarised thermal dust emission (red) and CMB polarisation (cyan). The dashed black lines indicate the sum of foregrounds evaluated over three different masks with $f_{sky} = 0.83$, 0.52, and 0.27. The widths of the synchrotron and thermal dust bands are defined by the largest and smallest sky coverages. Taken from [76].

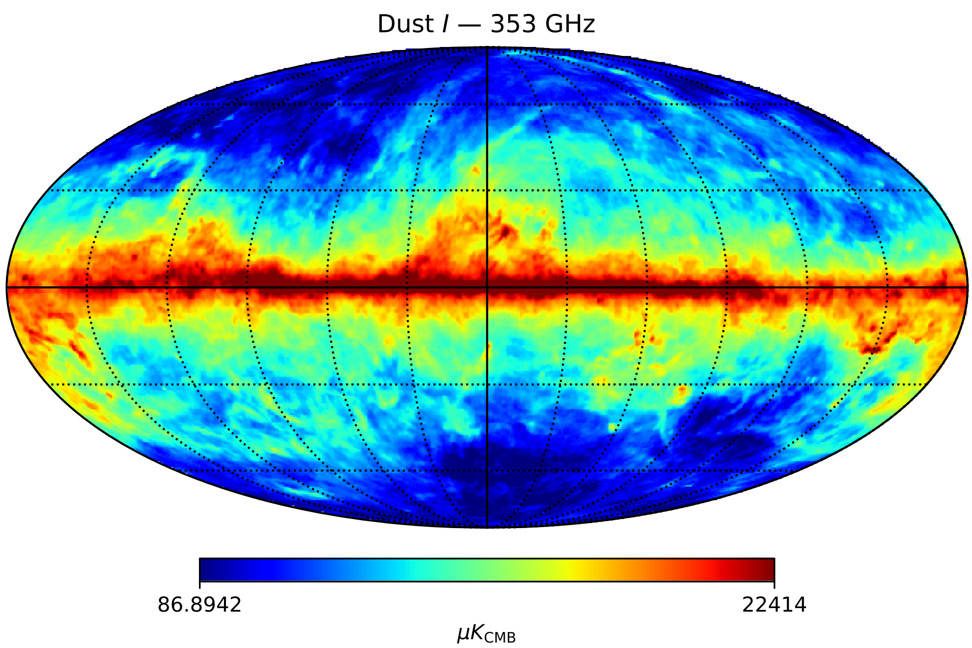

Figure 1.20: Thermal dust intensity map at 353 GHz from the Planck Commander analysis [76].

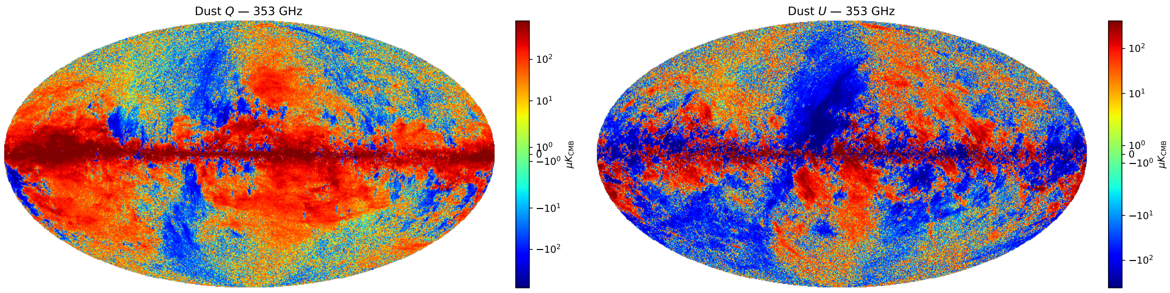

Figure 1.21: Thermal dust polarisation maps at 353 GHz from the Planck Commander analysis [76]. (left) Stokes Q map. (right) Stokes U map.

The simplest synchrotron model, called s_0, asserts that its emission spectrum follows a power law in frequency space, which can be modelled as:

$$I_{\nu,s} \propto \nu^{\beta_s}, \tag{1.37}$$

where β_s is the synchrotron spectral index, typically ranging between -3.2 and -2.5 depending on the region of the sky.

At frequencies below $\sim$ 70 GHz, synchrotron emission dominates over CMB, as shown in Figure 1.19. Its spectral energy distribution (SED) is commonly parameterised as:

$$f_s(\nu) = A_s \left(\frac{\nu}{\nu_{0,s}}\right)^{\beta_s}, \tag{1.38}$$

where A_s is the amplitude at reference frequency $\nu_{0,s}$ and β_s is the synchrotron spectral index. In contrast to thermal dust emission, synchrotron radiation does not depend on a temperature parameter, but exhibits significant spatial variations of β_s due to the distribution and energy spectrum of cosmic-ray electrons.

Figures 1.22 and 1.23 show respectively intensity and polarisation synchrotron maps at 30 GHz, as measured by Planck and reconstructed using the *Commander* framework [76].

1.4.4 Impact of astrophysical foregrounds on CMB polarisation

Astrophysical foregrounds constitute a major limitation for the observation of CMB polarisation. As shown in Figure 1.19, synchrotron emission dominates at low frequencies (below $\sim$ 70 GHz), while thermal dust emission dominates at high frequencies (above $\sim$ 100 GHz). The CMB signal lies between these two regimes, defining an optimal observation window around $\sim$ 70-150 GHz.

The impact of these foregrounds on CMB polarisation measurements is further illustrated in Figure 1.24, which compares their angular power spectra to that of the CMB. While the E-mode signal remains detectable, foreground contamination is particularly critical for B-modes, whose amplitude is significantly lower. In particular, thermal dust emission can exceed the expected primordial B-mode signal over a wide range of angular scales, even for relatively large values of the tensor-to-scalar ratio r.

These results highlight that foreground emission is a major obstacle to the detection of primordial CMB B-modes. However, dust and synchrotron emissions exhibit spectral

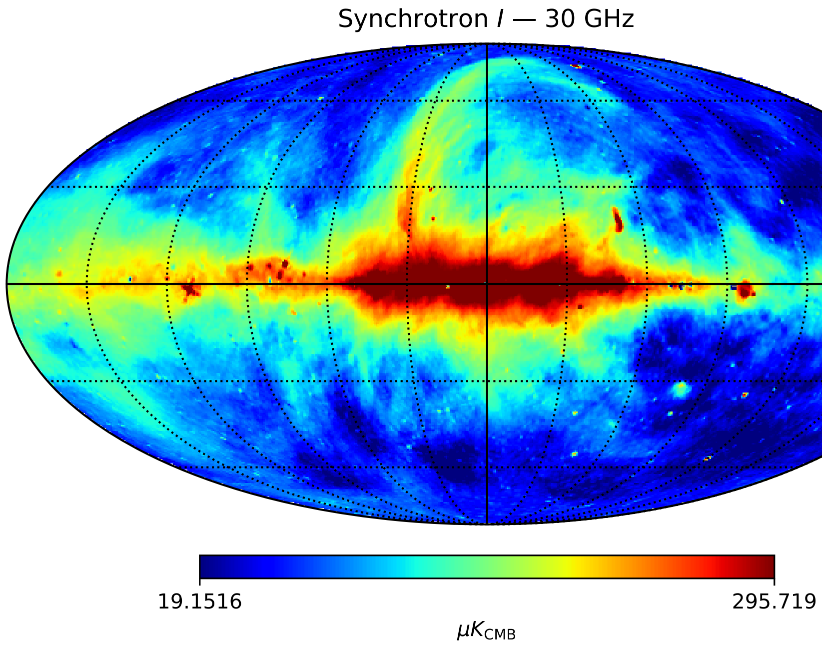

Figure 1.22: Synchrotron intensity map at 30 GHz from the Planck Commander analysis [76].

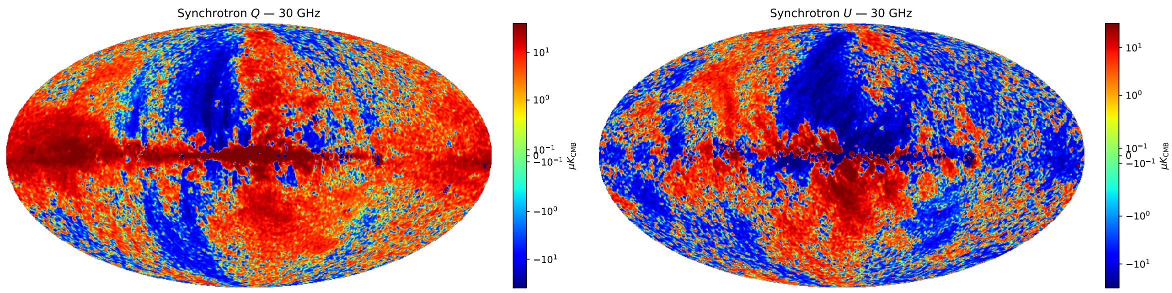

Figure 1.23: Synchrotron polarisation maps at 30 GHz from the Planck Commander analysis [76]. (left) Stokes Q map. (right) Stokes U map.

behaviours that differ from that of the CMB, as described by Eqs. 1.36 and 1.38. This difference provides a way to disentangle the various contributions using multi-frequency observations.

This principle forms the basis of *component separation* methods, which aim at reconstructing the CMB signal from contaminated sky observations. These techniques will be introduced and developed in the following Chapters: 5, 6 and 10.

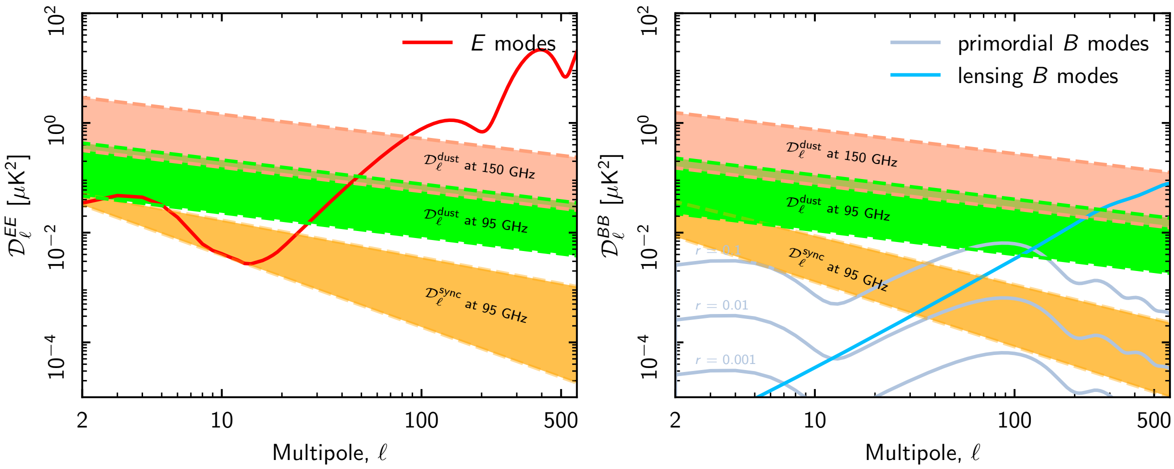

Figure 1.24: (left) Dust (95 and 150 GHz) and synchrotron (95 GHz) E-modes power versus CMB. (right) Dust (95 and 150 GHz) and synchrotron (95 GHz) B-modes power versus CMB ($r = 0.1, 0.01, 0.001$). The coloured bands show the range of power measured from the smallest to the largest sky regions in Planck analysis. Taken from [75].

1.4.5 Atmosphere

In addition to astrophysical foregrounds, ground-based CMB experiments must contend with contamination from the Earth's atmosphere. The atmosphere is not a transparent medium at microwave frequencies and both absorbs and emits radiation, thereby contributing to the measured signal.

In particular, atmospheric brightness is strongly affected by the amount of water vapour, as well as by temperature and pressure, as illustrated in Figure 1.25. The atmospheric spectrum at CMB frequencies is dominated by absorption lines from dioxygen (O_2) and water vapour (H_2O). Between these lines, partially transparent observational windows exist, notably around 90 GHz, 150 GHz, and 220 GHz.

The overall emission level within these windows depends on the amount of water vapour present in the atmosphere, commonly quantified by the Precipitable Water Vapour (PWV). PWV corresponds to the total column-integrated water vapour content, expressed as the equivalent height of liquid water obtained upon condensation. It is defined as:

$$\mathrm{PWV} = \frac{1}{\rho_w} \int_0^\infty \rho_v(z)\, dz, \tag{1.39}$$

where ρ_w is the density of liquid water ($1000\,\mathrm{kg/m^3}$), and $\rho_v(z)$ is the water vapour density at altitude z.

Low values of PWV are essential for high-sensitivity observations, as they reduce both atmospheric emission and absorption. For this reason, CMB experiments are typically located at high-altitude, dry sites.

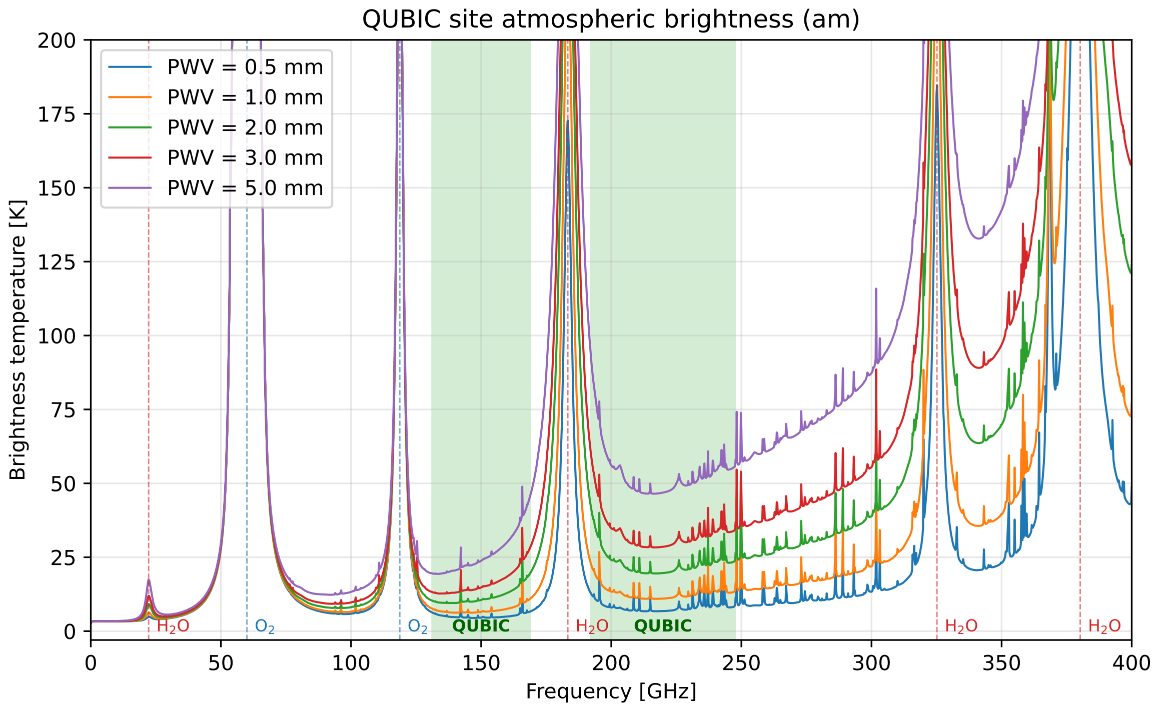

Figure 1.25: Zenith brightness temperature as a function of frequency for different PWV values. The two observational bands and the main absorption lines and their associated molecules are indicated. Simulated using the *am* software [78].

In addition to its average emission, the atmosphere is a turbulent medium, leading to fluctuations in the observed signal along the line of sight and during telescope scanning, leading to spatial and temporal correlations in data. These fluctuations are commonly described by Kolmogorov turbulence, characterised by a three-dimensional power spectrum:

$$P(\vec{k}) \propto |\vec{k}|^{-11/3}, \tag{1.40}$$

where $\vec{k}$ is the spatial wave vector [79, 80].

Atmospheric emission and fluctuations constitute one of the main limitations for next-generation ground-based CMB experiments. Developing efficient mitigation techniques is therefore essential, and this will be the focus of Chapter 11.

1.5 Past & Future Experiments

We focus here on CMB-dedicated experiments. Other cosmological probes (e.g. galaxy surveys such as Euclid or LSST) provide complementary constraints and will be briefly discussed at the end of this section. We present here a selection of landmark CMB experiments that have shaped our current understanding of CMB physics. While not exhaustive, it highlights the most influential missions and observatories across successive generations.

1.5.1 Past Experiments

COBE

The Cosmic Background Explorer (COBE, 1989–1993) [81] was the first satellite dedicated to precise measurements of the CMB. It carried three instruments: FIRAS, DMR, and DIRBE.

Its main contributions are:

- Measurement of the CMB spectrum as an almost perfect blackbody, confirming its thermal origin [82].
- First detection of large-scale temperature anisotropies at the level of $\Delta T/T \sim 10^{-5}$ [55].

COBE established the observational foundation of modern cosmology and demonstrated that primordial fluctuations exist, leading to the obtention of Nobel Prize in 2006[8].

WMAP

The Wilkinson Microwave Anisotropy Probe (WMAP, 2001–2010) [83] provided the first high-resolution full-sky maps of CMB temperature anisotropies.

Its main contributions include:

- Precise determination of the standard cosmological parameters.
- Strong evidence for a flat Universe dominated by dark energy and dark matter.
- First significant measurements of large-scale polarisation.

Planck

The Planck satellite (2009–2013) [84, 18] represents the most precise full-sky CMB experiment to date.

Its main contributions are:

- Highest precision measurements of temperature anisotropies (at cosmic variance limit on a wide range of angular scales) and high signal-to-noise polarisation maps.
- First full-sky, high-precision measurements of CMB E-mode polarisation over a broad range of angular scales.
- Measurement of lensing-induced B-modes through the CMB lensing potential reconstruction.

[8] `https://www.nobelprize.org/prizes/physics/2006/summary/`

- Tight constraints on ΛCDM parameters and extensions.

Planck provides the current reference dataset for cosmology Λ-CDM model.

1.5.2 Current Experiments

ACT

The Atacama Cosmology Telescope (ACT) [85] is a ground-based experiment located in Chile, optimised for high angular resolution observations.

Its main contributions include:

- Measurements of small-scale anisotropies.
- High-significance detection of CMB lensing.
- Constraints on structure formation and neutrino properties.

SPT

The South Pole Telescope (SPT) [86] is a ground-based experiment operating at the South Pole.

Its main contributions are:

- Detection of galaxy clusters via the Sunyaev–Zel'dovich effect.
- Measurements of small-scale power spectrum and lensing.
- Constraints on dark energy and neutrino mass.

BICEP/Keck Array

The BICEP/Keck Array experiments [87] focus on large-scale CMB polarisation, particularly B-modes.

Their main contributions include:

- Leading constraints on the tensor-to-scalar ratio r.
- Development of advanced foreground (dust) separation techniques.

They play a central role in the search for primordial gravitational waves.

Simons Observatory

The Simons Observatory [88] is a new generation ground-based experiment in Chile, currently entering operation.

Its scientific goals include:

- Improved measurements of temperature and polarisation over a wide range of scales.
- Constraints on inflation through r.
- Improved bounds on neutrino mass and relativistic species.

1.5.3 Future Experiments

LiteBIRD

LiteBIRD [89] is a planned space mission led by JAXA, dedicated to large-scale CMB polarisation.

Its primary objective is:

- Detection (or stringent constraint) of primordial B-modes.

By observing from space, LiteBIRD will minimize atmospheric contamination and systematics.

FOSSIL

FOSSIL (FTS fOr CMB Spectral diStortIon expLoration) [53] is a space mission proposed to the ESA M7 call, designed to survey the sky from 30 to 2000 GHz and improve upon COBE/FIRAS sensitivity by three orders of magnitude.

Its science objectives span a wide range of fundamental questions:

- Probing early Universe physics and inflation.
- Constraining black-hole formation scenarios and the nature of dark matter.
- Characterising structure formation and the blackbody nature of the CMB.
- Exploring recombination physics at last scattering and the end of the reionisation epoch.

The FOSSIL concept builds on the heritage of COBE/FIRAS and previous proposals (PIXIE, PRISTINE), and was submitted by a consortium of more than 40 institutions across 15 countries. By targeting CMB spectral distortions, tiny departures of the CMB energy spectrum from a perfect blackbody, FOSSIL offers a fundamental probe of the cosmic thermal history.

Other Cosmological Probes: While CMB experiments provide some of the strongest constraints on cosmology, future progress will rely on their combination with large-scale structure surveys (e.g. Euclid, LSST, DESI), enabling joint analyses of geometry, growth of structure, and fundamental physics.

Despite the remarkable progress achieved by current and upcoming CMB experiments, the measurement of primordial B-modes remains extremely challenging. The main limiting factors are the control of instrumental systematics and the accurate separation of astrophysical foregrounds, particularly Galactic dust emission.

While most existing experiments rely on imaging techniques, these challenges motivate the exploration of alternative instrumental approaches that can provide improved control over systematics and enhanced robustness to foreground contamination.

QUBIC

The QUBIC (Q & U Bolometric Interferometer for Cosmology) experiment [90] is a ground-based instrument designed to measure CMB polarisation using the novel concept of *bolometric interferometry*.

This approach combines the high sensitivity of bolometric detectors with the systematic control advantages of interferometry. Instead of forming images directly on the sky, QUBIC measures interference patterns produced by an array of horns, allowing for:

- Intrinsic mitigation of certain instrumental systematics through self-calibration.
- Improved control of foreground contamination via spectral imaging.
- Enhanced control over atmosphere (as explored for the first time in this thesis).

QUBIC is therefore complementary to traditional imaging experiments such as ACT, SPT, BICEP/Keck and Simons Observatory, and represents an alternative strategy in the quest for primordial B-mode detection.

A Technological Demonstrator (TD) is installed at the Alto Chorrillos site in Argentina, and is currently in its commissioning phase.

QUBIC constitutes the central experimental framework of this thesis. The following chapter is therefore dedicated to a detailed presentation of the collaboration, the instrument, and its scientific objectives.

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